

	Work Instruction	Doc. Revision	01
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This procedure is applicable to below models:

V810 STD	☒	S1	☒	S2
V810 XXL	☒	S1	☒	S2
V810 S2EX	☒			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	12 Apr 2016	First Release	MUHAMMED SHAMEEM
01	2 Oct 2016	Template Changes & Content Update	ONG KEAN SHENG

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Pre-Requisites step

1. May refer to Table 1 for software requirement for system types.

System Type	Software requirement
Standard V810	5.8 & above
XXL V810	5.8 & above
V810 S2EX	5.8 & above

Table 1 Software requirement

2. May refer to Table 2 for desire adjustment method.

System Type	Machine Serial Number	Adjustment method	Tuning Procedure
Standard V810	MY10133110 & below	Manual Load Jig	Automated PSP 2 Adjustment Procedure (Manual Load Jig)
	MY10133111 & above	On Rail Jig	Automated PSP 2 Adjustment Procedure (On Rail Jig)
XXL V810	All	On Rail Jig	Automated XXL PSP 2 Adjustment Procedure (On Rail Jig)
V810 S2EX	All	On Rail Jig	Automated S2EX PSP 2 Adjustment Procedure (On Rail Jig)

Table 2 Adjustment Method

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3. If particular machine is within “On Rail Jig Adjustment method”, will require visual check at On rail jig .May to Figure 1 and Figure 2 for further reference.

Remarks: On Rail Jig which illustrated at below would be the latest version.

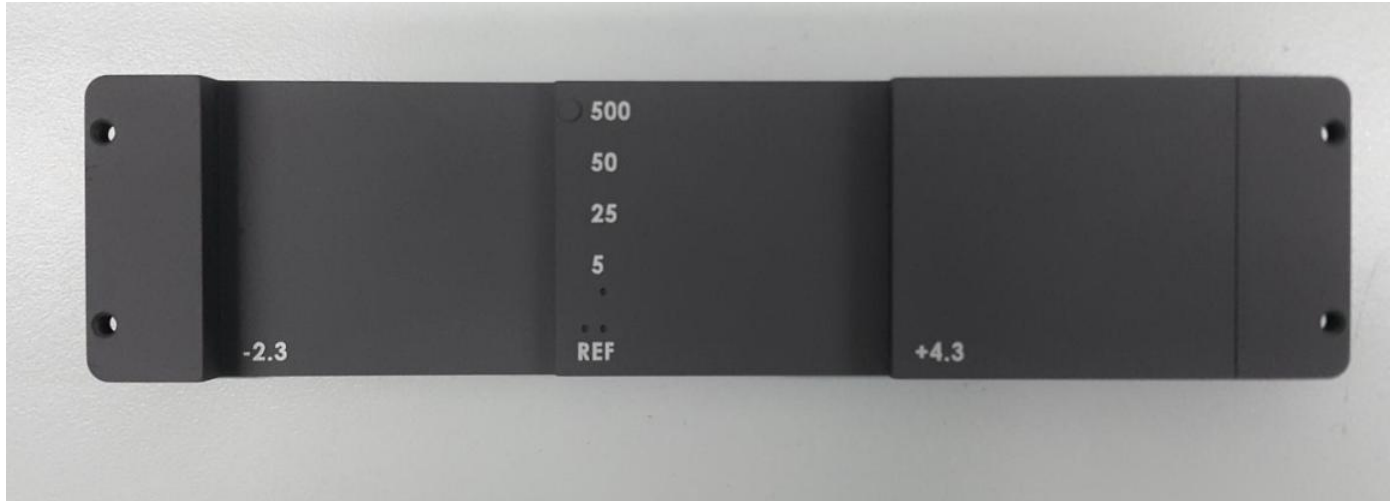


Figure 1 Top View of the On Rail Jig (Latest version)

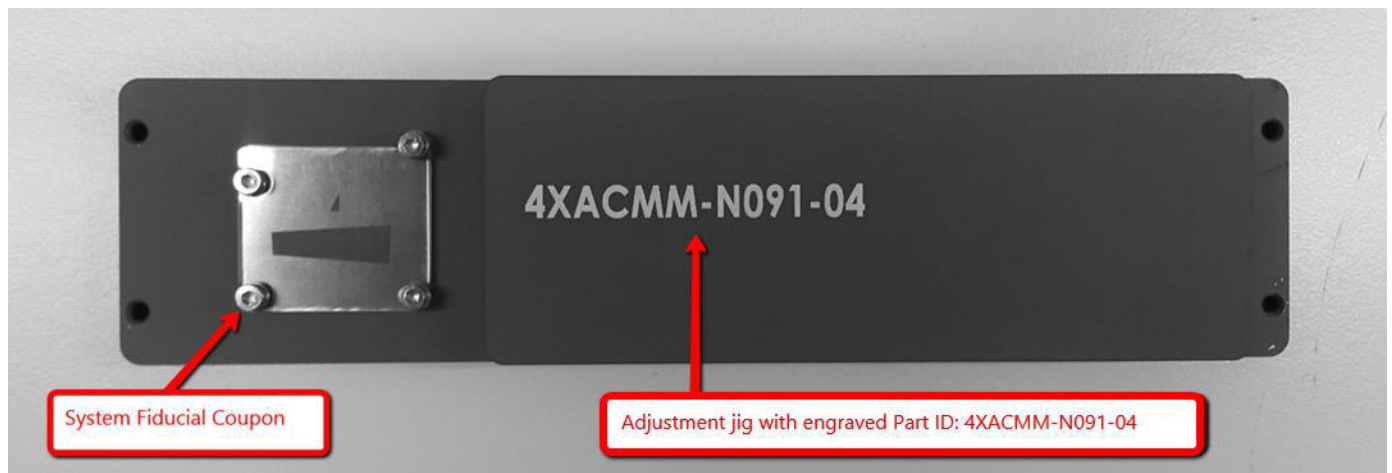


Figure 2 Bottom View of the On Rail Jig (Latest version)

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Adjustment Steps:

1. Run 1 loop of Confirmation, Diagnostics and Adjustment (CD&A) test. Once CD&A task complete, close V810 software.
2. Go to the following directory **C:\Program Files\AXI System\v810\5.0\calib** and delete the following files which listed in Table 3.

System Model	Directory	Files Required to be deleted
All Model	C:\Program Files\AXI System\v810\5.0\calib	i.throughput.hardware.calib.binary ii. throughput.psp.calib.binary
	C:\Program Files\AXI System\v810\5.0\log\calib\OpticalCalibration	All .png images
	C:\Program Files\AXI System\v810\5.0\log\calib\OpticalSystemOfsetAdjustment	All .png images

Table 3 Previous Adjustment Files

3. Open V810 GUI and run 1 loop of Confirmation, Diagnostics and Adjustment (CD&A)
4. Open Software Config by refer below path where depend Equipment Model.

Model	Path
Strd Series One & Series 2	C:\Program Files\AXI System\v810\5.0\Config
XXL Series One & Series 2	C:\Program Files\AXI System\v810\5.0\xxlConfig
S2EX	C:\Program Files\AXI System\v810\5.0\s2exConfig

Change **enablePsp=false** to **enablePsp=true** as shown in the Figure 3

Save and close the software.config after change the keys.

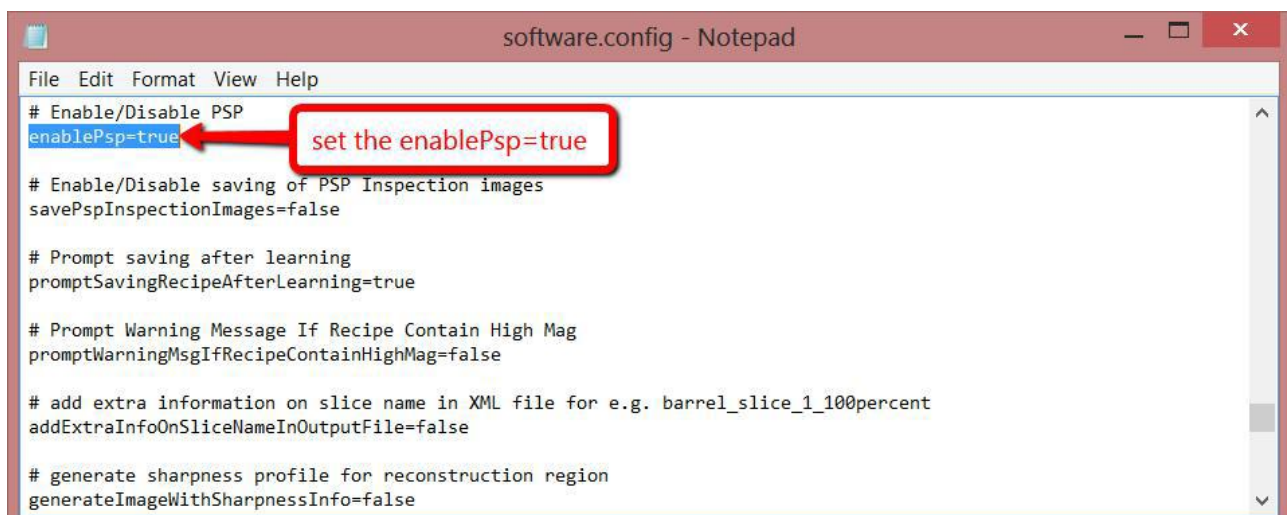


Figure 3 Software.config

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5. Open Hardware Config by refer below path where depend Equipment Model.

Model	Path
Strd Series One & Series 2	C:\Program Files\AXI System\v810\5.0\Config
XXL Series One & Series 2	C:\Program Files\AXI System\v810\5.0\xxlConfig
S2EX	C:\Program Files\AXI System\v810\5.0\s2exConfig

6. Set the following keys at the hardware.config mentioned below (refer to Figure 4):

pspModuleVersion=2

pspCalibrationVersion=3

pspSetting=1

opticalCameraModel=axiOpticalCameraType2

Save and close the hardware.config.

```

File Edit Format View Help
## PSP Module settings
##
## Version 1 = Legacy PSP module (with DLP Projector)
## Version 2 = New PSP module (with EBLE projector)
#####
pspModuleVersion=2
#####
## PSP Calibration settings
##
## Version 1 = Manual-loaded PSP calibration jig
## Version 2 = XXL On-Stage Calibration jig (old jig with 5 plane)
## Version 3 = On-Stage Calibration jig (new jig with 3 plane) - For Std and XXL
## Version 4 = New Manual-loaded PSP calibration jig (with 3 plane) - For Std Only
#####
pspCalibrationVersion=3
#####
## PSP Module Connection settings
##
## 1 = Connect to Both Front and Rear Module
## 2 = Connect to Front Module Only
## 3 = Connect to Rear Module Only
#####
pspSetting=1
#####
## PSP On-Stage Jig Object Plane Z-Height (in micron)
#####
pspOnStageJigObjectPlaneZHeightInMicron=4300
#####
## Optical camera settings
#####
opticalCameraModel=axiOpticalCameraType2
opticalCameraLoggingEnabled=false
#####
## Micro-Controller settings
#####
pspMicroControllerSerialPortName=COM4
#####
## Ethernet-Based Light Engine
##
## Supported Setting
## 1) Current (mA) : 350 (default), 495, 635, 770
## 2) Period : 20 (default), 30, 40, 50
#####
ethernetBasedLightEngineHostIpAddress=123.123.123.200
ethernetBasedLightEngineCurrentSettingMicroAmps=350
ethernetBasedLightEnginePeriod=20

```

Figure 4 Hardware.config

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7. Open V810 GUI and go to the and run 1 loop of Confirmation, Diagnostics and Adjustment (CD&A)
8. At **Confirmation, Diagnostics and Adjustment (CD&A)**, under **Manual Optical Adjustments**, right click **Optical System Fiducial Adjustment** and select **Run test without dependent tests** as shown in the Figure 5.

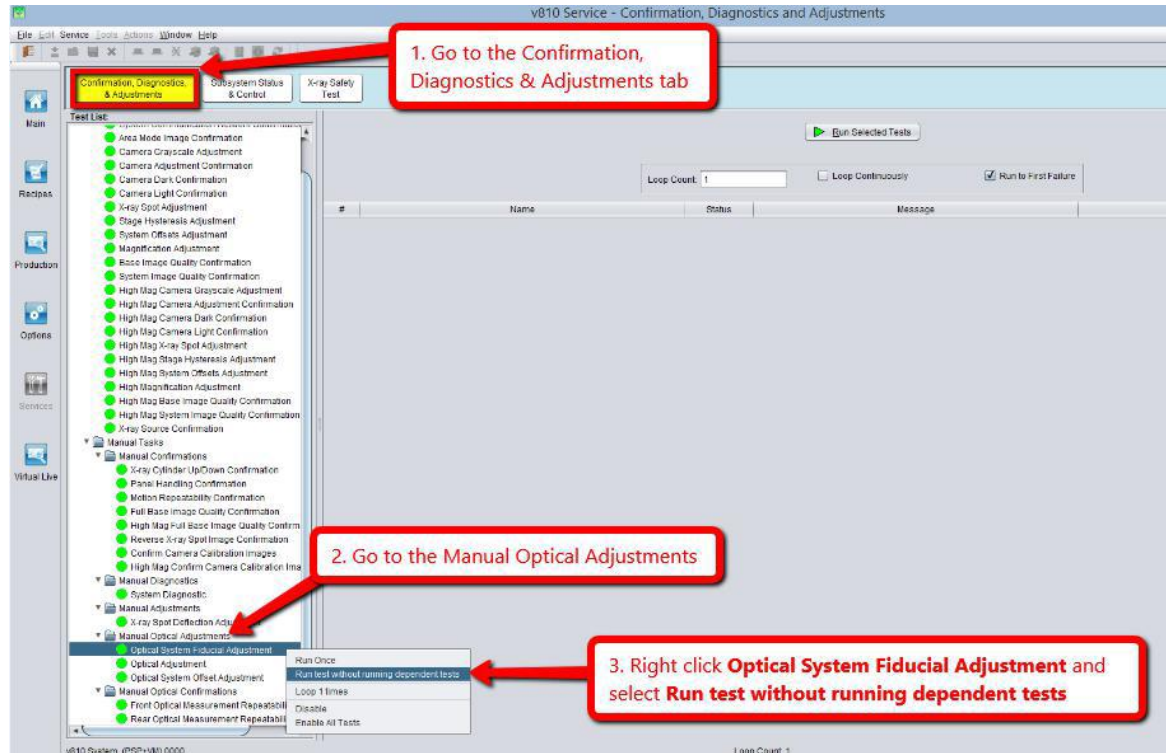


Figure 5 Optical System Fiducial Adjustment

9. XY stage will move to Front PSP Module and the image will be generated as shown in Figure 6.

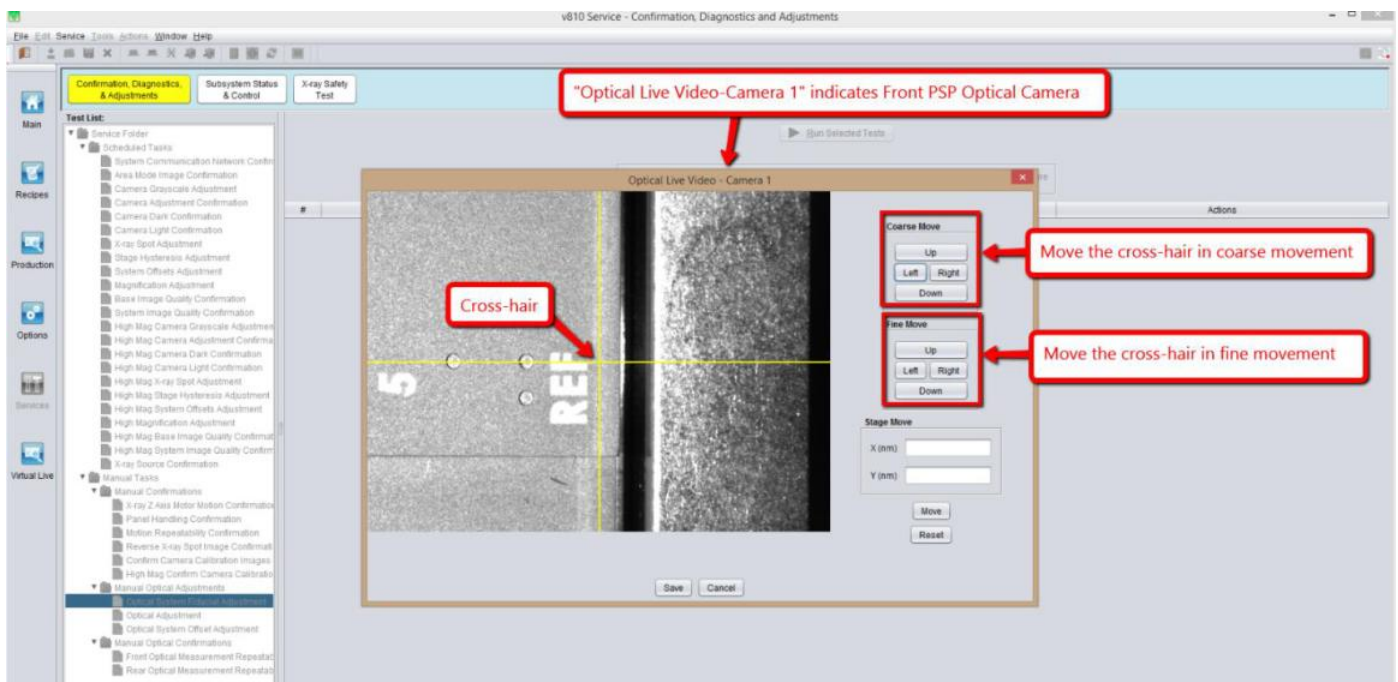


Figure 6 Front PSP Optical Camera

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10. Move cross-hair to particular location as Figure 7.

Click save.

Remarks: Cross hair require to be within in middle of the circle as Figure 8.

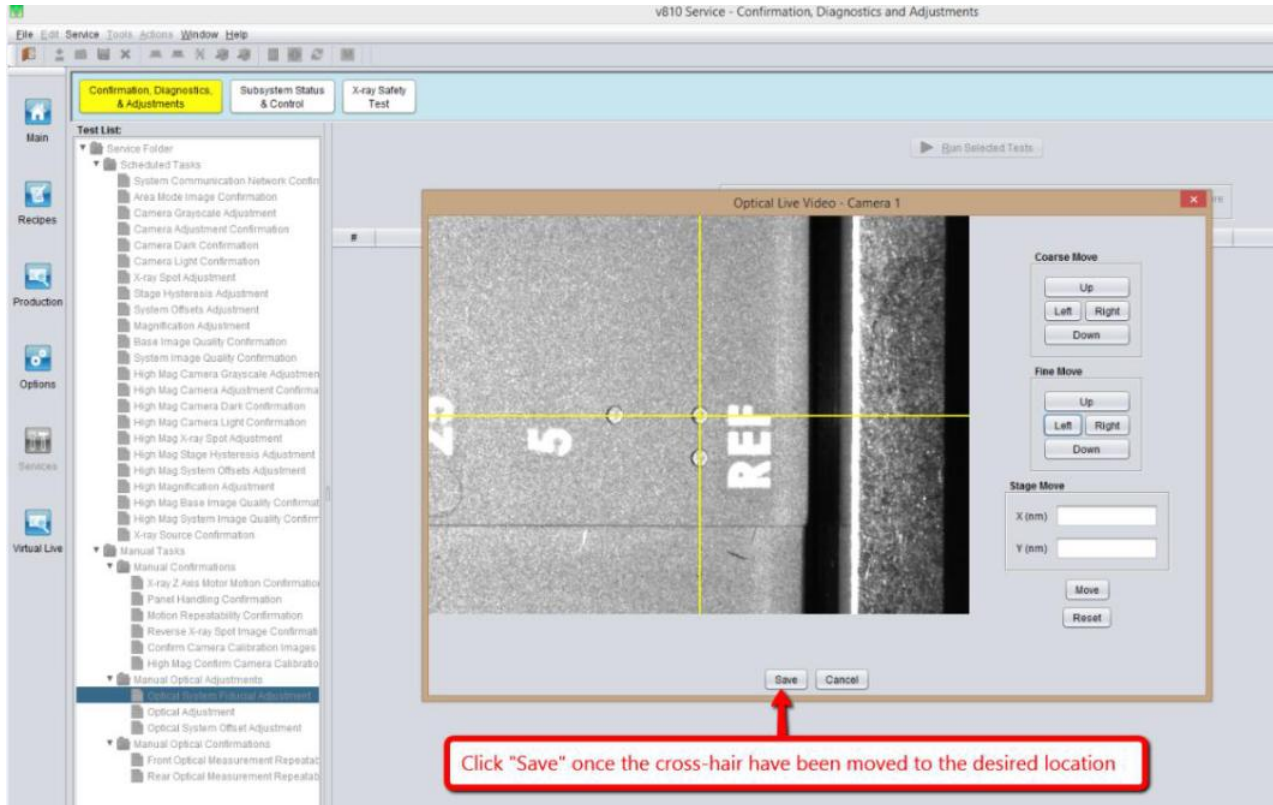


Figure 7 Front PSP Module Adjustment

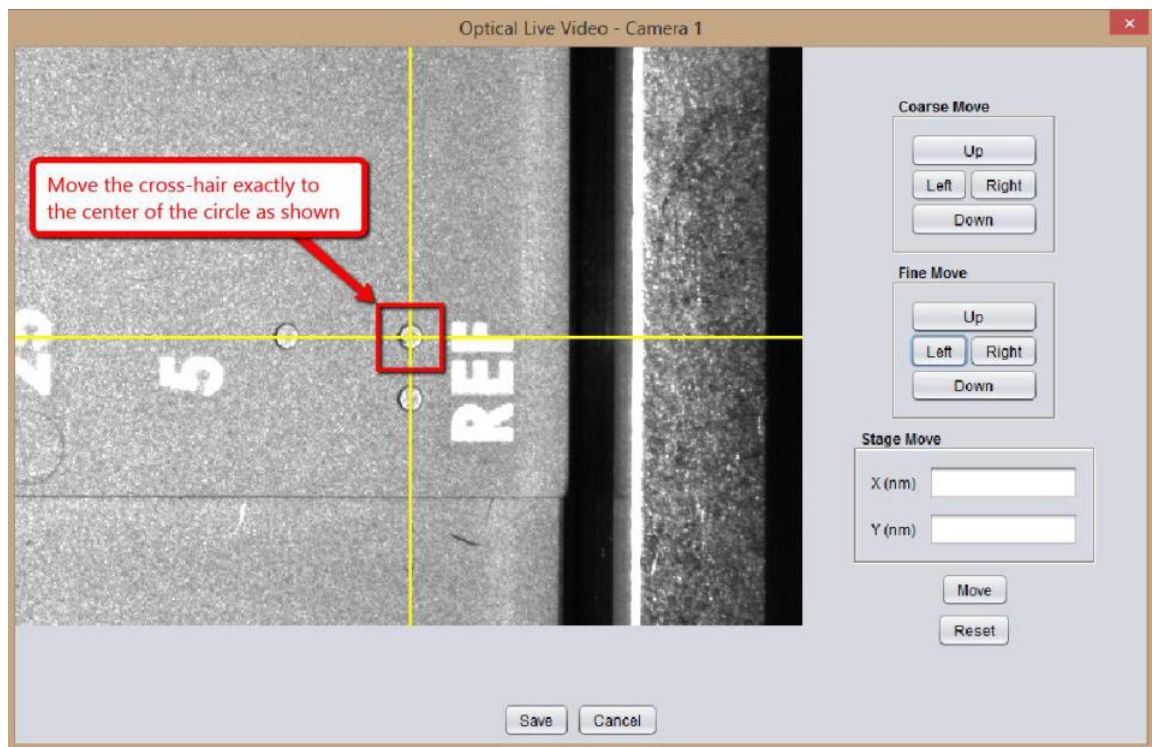


Figure 8 Exact point to select the cross hair

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11. Next, Optical Live view camera 2(Rear PSP Camera) will be shown as Figure 9.

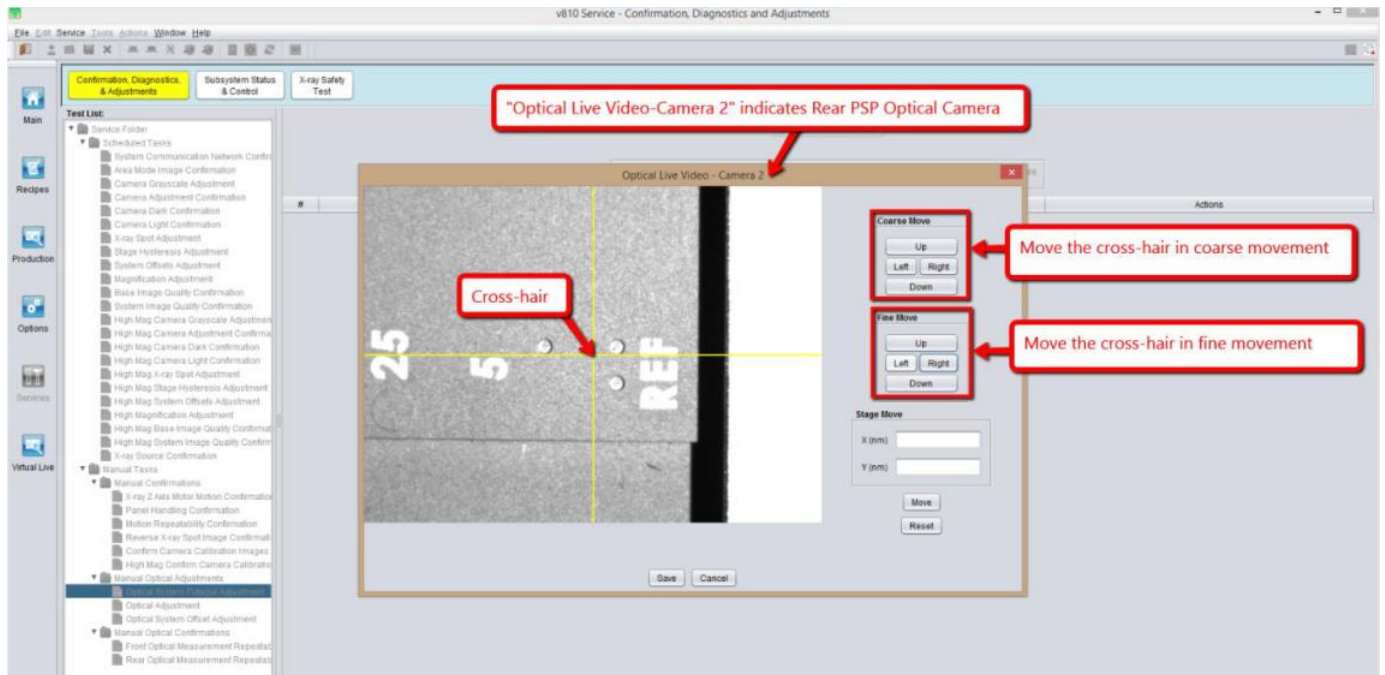


Figure 9 Rear PSP Optical Camera

12. Move the cross-hair to particular location as Figure 10.

Click save.

Remarks: Cross hair require to be within in middle of the circle as Figure 10.

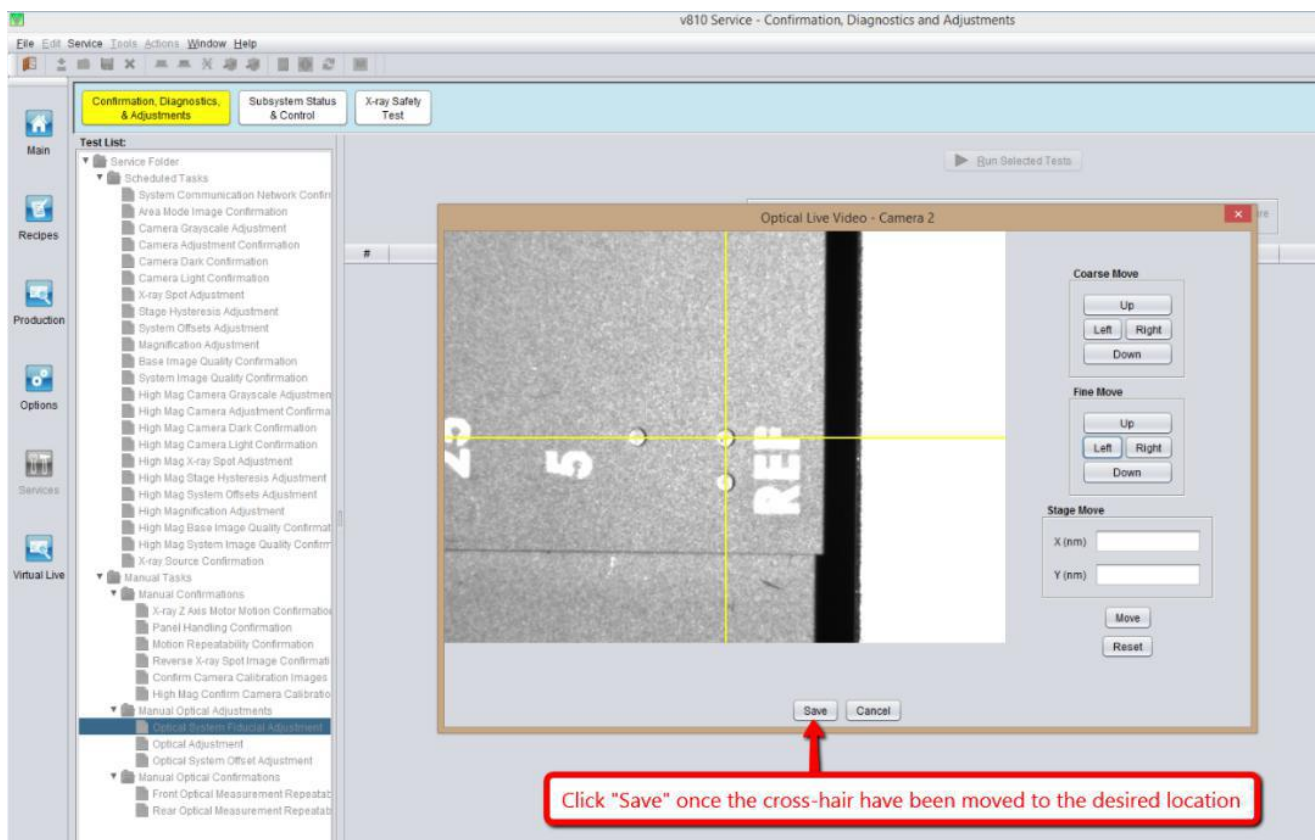


Figure 10 Rear PSP Fiducial Point selection

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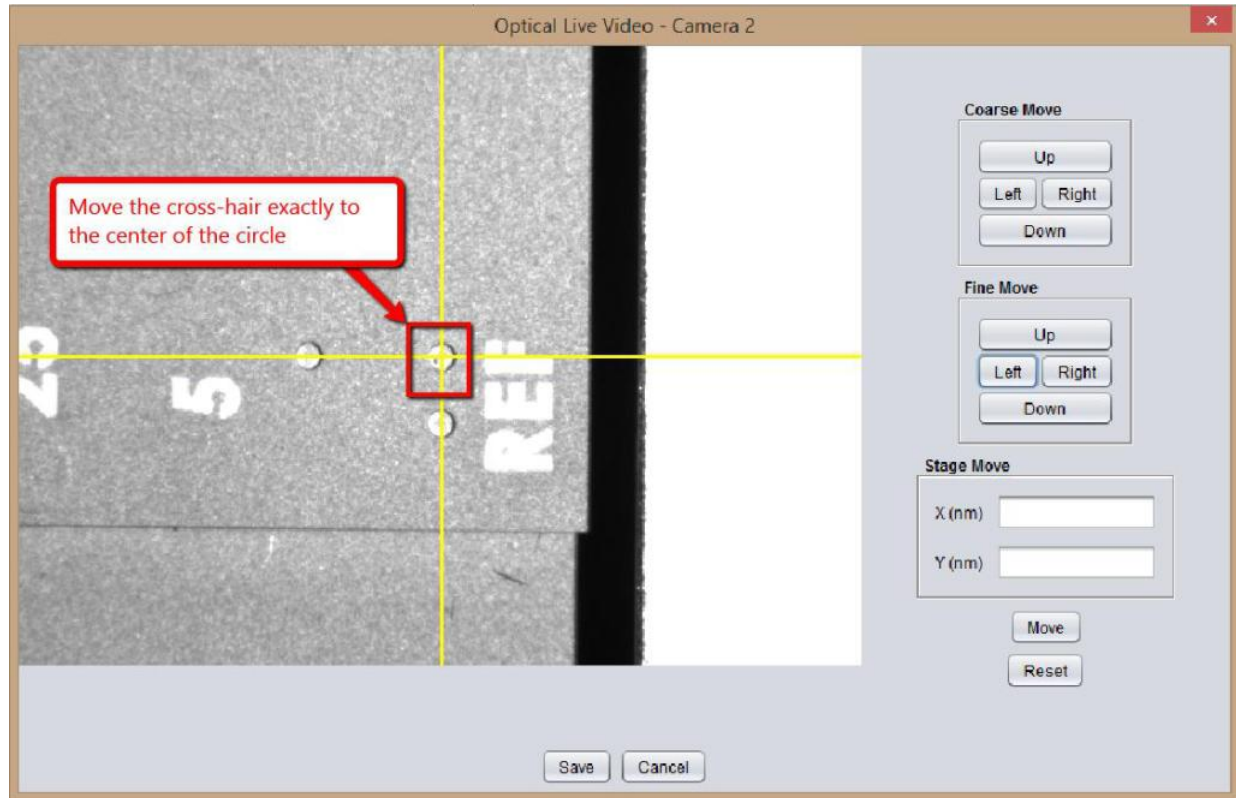


Figure 11 Exact point for cross hair selection

13. Optical System Fiducial Adjustment result will be illustrate as Figure 12.

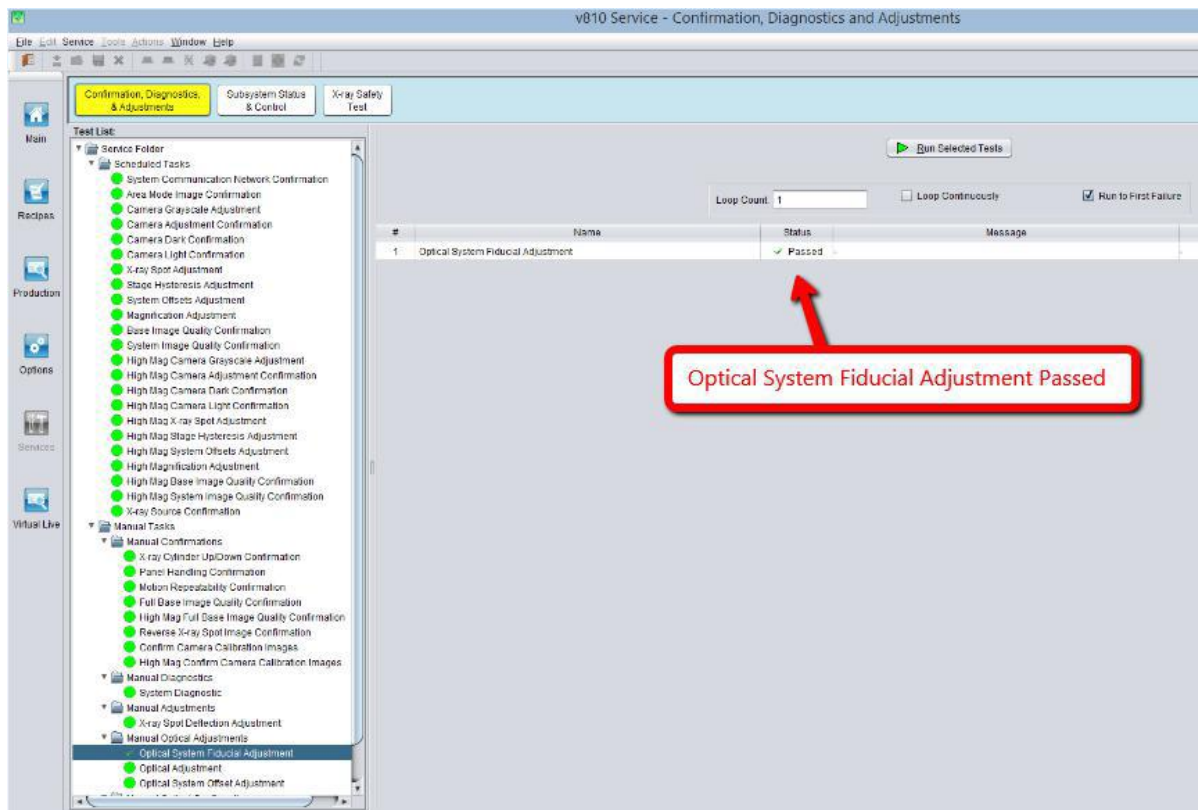


Figure 12 Optical System Fiducial Adjustment Passed

14. At **Confirmation, Diagnostics and Adjustment (CD&A)** , under the **Manual Optical Adjustments**, right click **Optical Adjustment** and select **Run test without dependent tests** as shown in the Figure 13.

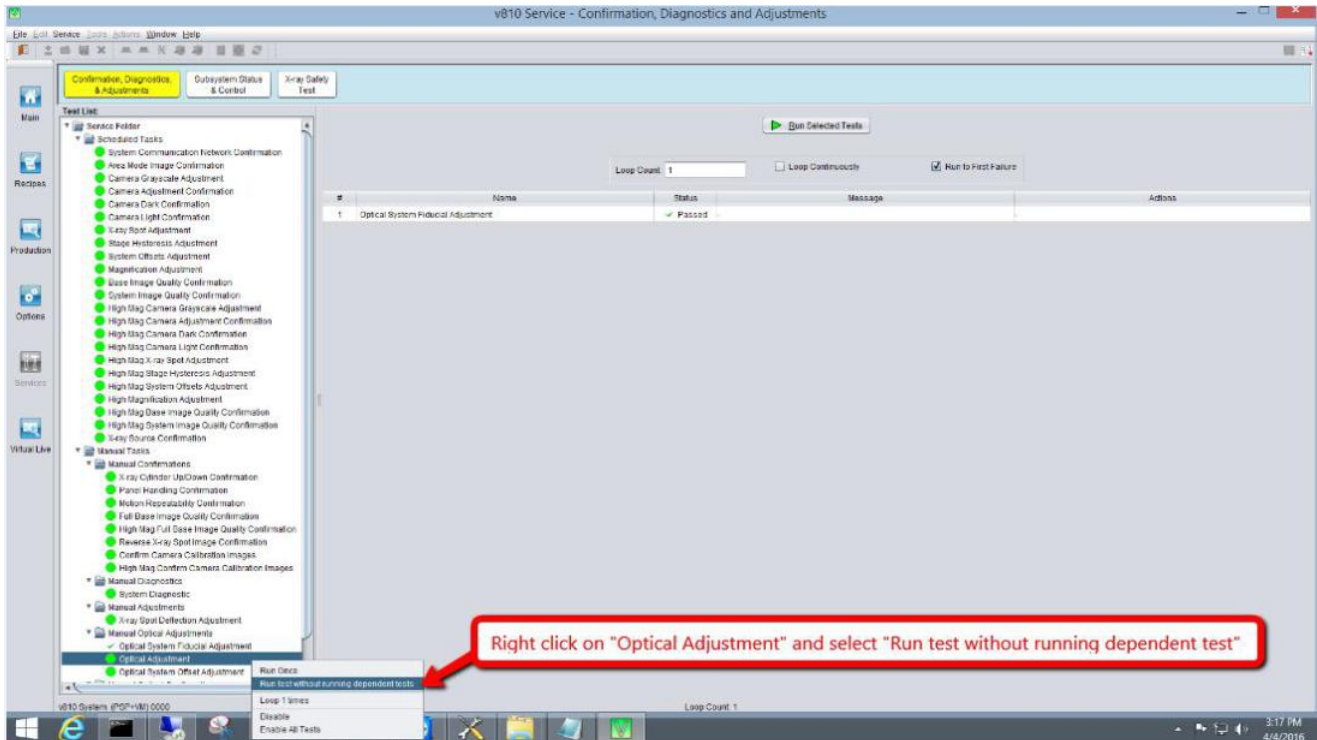


Figure 13 Optical Adjustment

15. Optical Adjustment result will be illustrate as Figure 14.

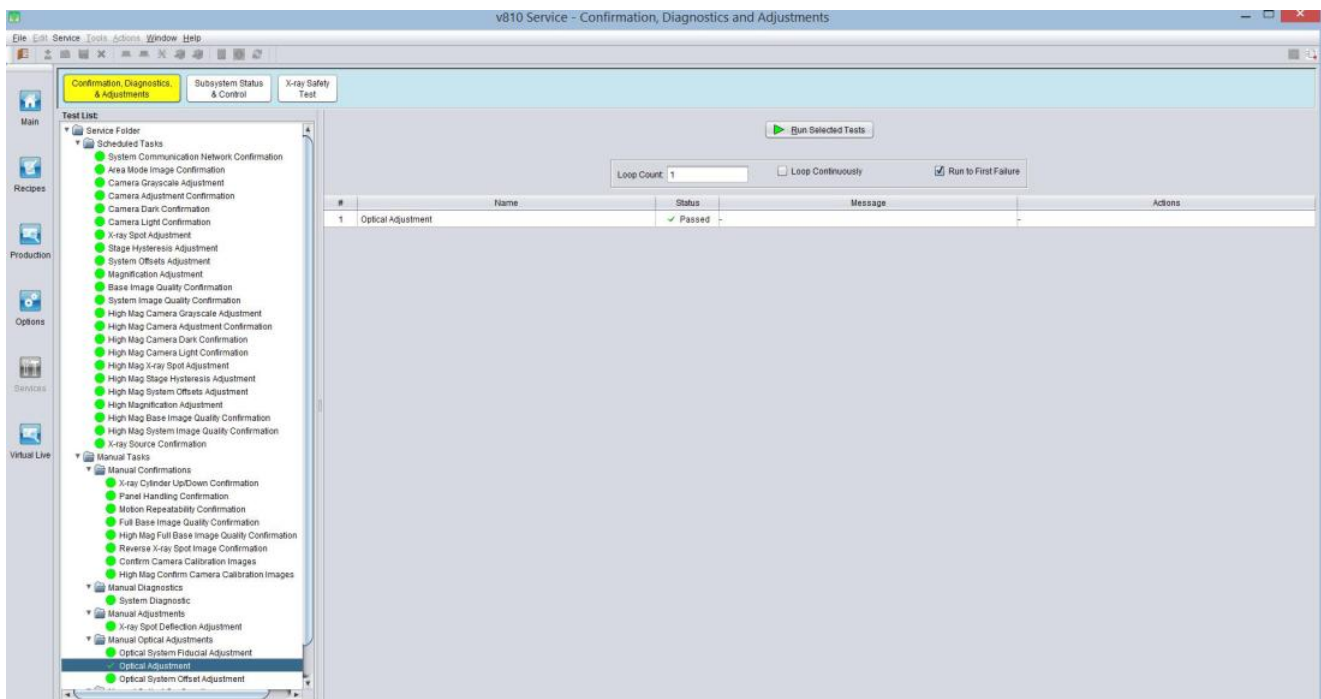


Figure 14 Optical Adjustment

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16. Go to the following directory

Directory: C:\Program Files\AXI System\v810\5.0\log\calib\OpticalCalibration

Total number of 32 images will be generated as shown in the Figure 15.

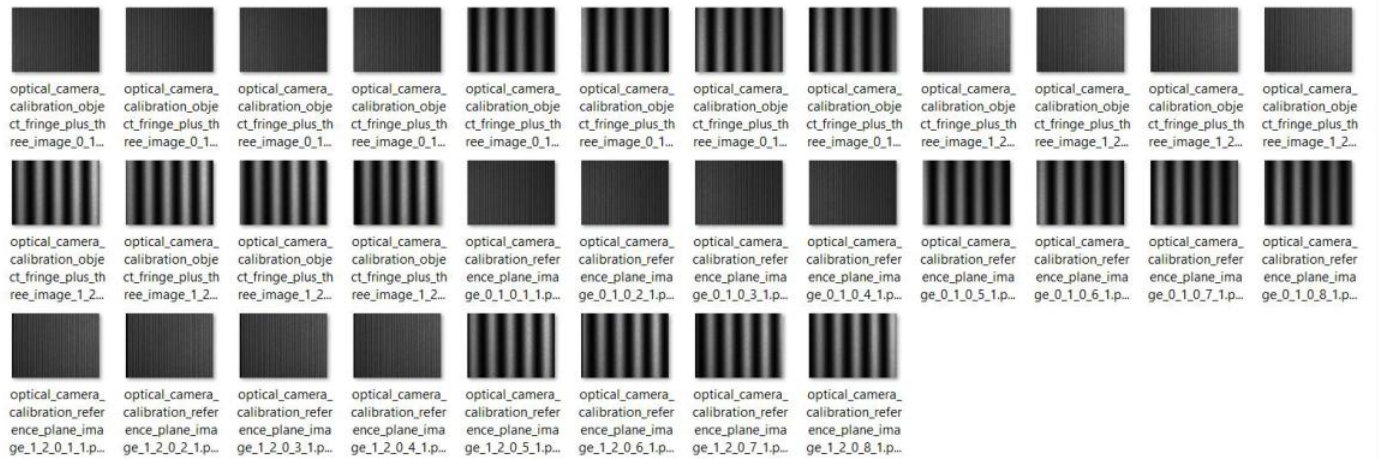
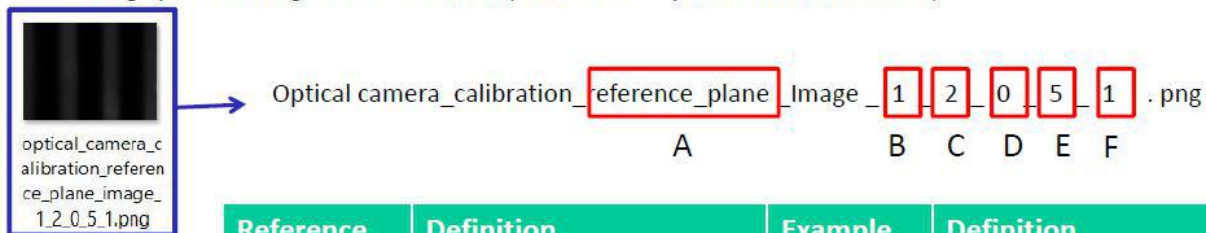


Figure 15 Optical Calibration Fringe Pattern

Additional reference for the Fringe Pattern Naming is as in Figure 16

Fringe pattern image name definition (Store under Optical calibration folder)



Reference	Definition	Example	Definition
A	Calibration image captured by using REF plane at PSP calibration jig	Reference_Plane	Calibration image captured by using REF plane at PSP calibration jig
B	Front Camera (id = 0) Rear Camera (id = 1)	1	Rear camera
C	The value will always 1 value higher than Camera Id. The values are 1 (front camera) and 2 (rear camera)	2	The value will always 1 value higher than Camera Id. Means If "B" is indicate "1" , this device ID will add on "1" at "B", so the camera device ID will be 2
D	Profile 1 (id = 0), Profile 2 (id = 1)	0	Default will set to "0" Means just using "REF" & "+3" plane at PSP jig only
E	For PSP 2.0, the sequence number starts from 1 to 8.	5	Capturing 5th of fringe pattern Sequence number starts from 1 to 8.
F	This is designed so that we can handle multiple run of calibration for averaging purpose. For now, it is default to 1.	1	Default set to "1" , means just capturing one image at one time only

Figure 16 Fringe pattern naming

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17. May refer to Figure 17 to 20 for failure detection.

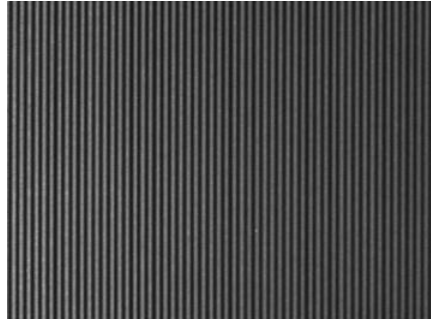


Figure 17 Good Images

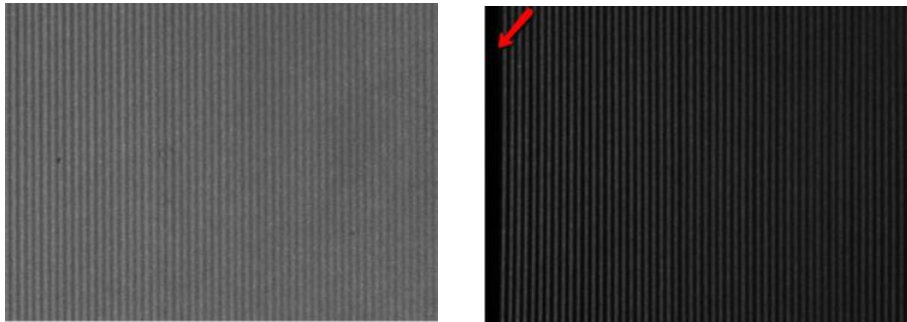


Figure 18 Bad Images

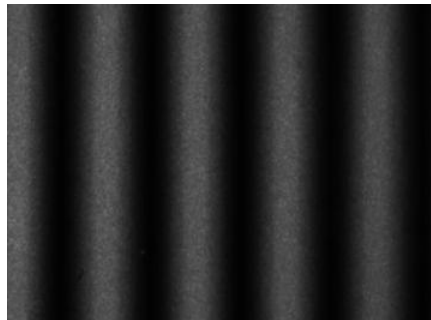


Figure 19 Good Images

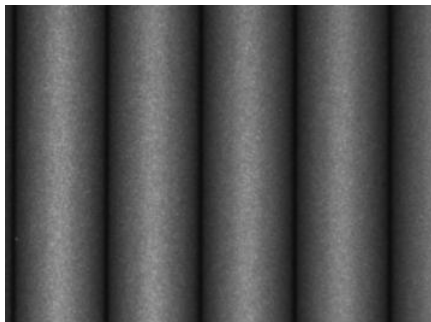


Figure 20 Bad Images

If adjustment is require,may refer to the procedure“AXI V810 Front Module Offline Adjustment Procedure (EQUIPMENT) ” and “AXI V810 PSP Rear Module Offline Adjustment Procedure (FOR EQUIPMENT) ”

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18. At **Confirmation, Diagnostics and Adjustment (CD&A)**, under the **Manual Optical Adjustments**, right click **Optical System Offset Adjustment** and select **Run test without dependent tests** as shown in the Figure 21.

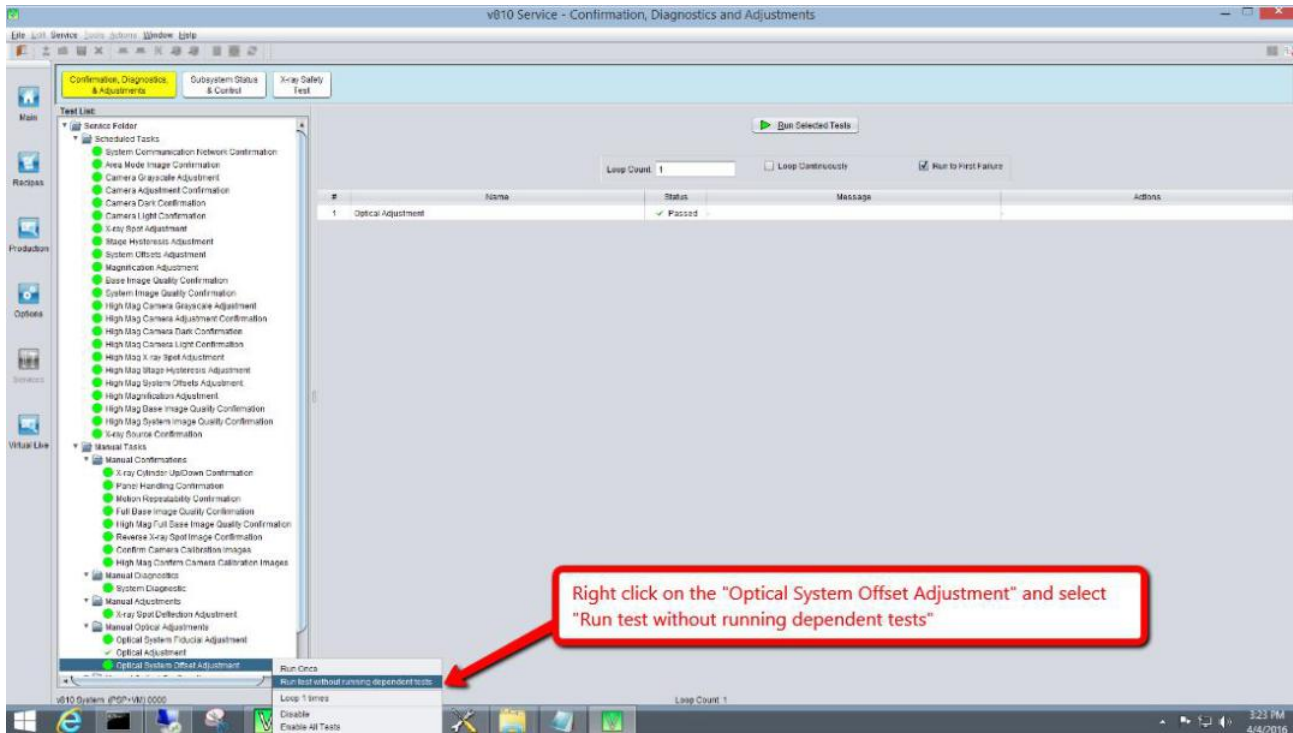


Figure 21 Optical System Offset Adjustment

19. Optical System Offset Adjustment result will be illustrated as Figure 22.

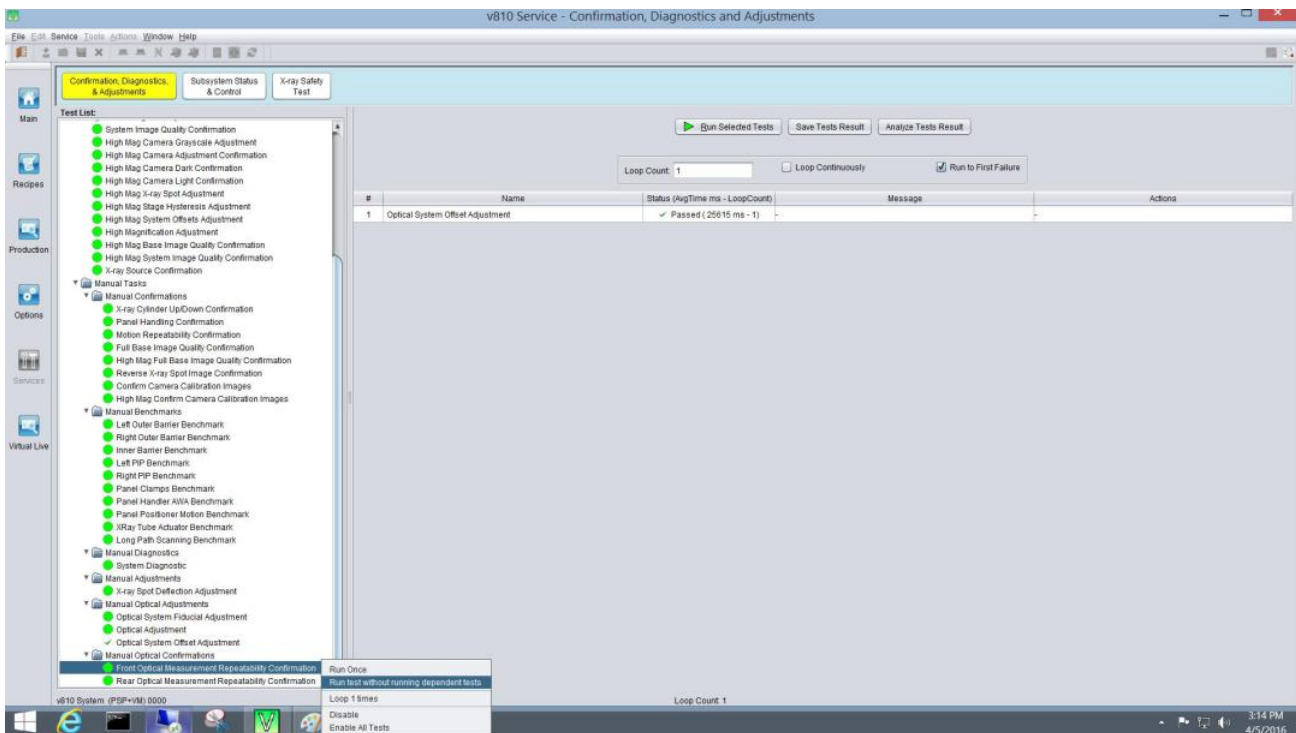


Figure 22 Optical System Offset Adjustment

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20. Go to the following folder.

Directory: C:\Program Files\AXI System\v810\5.0\log\calib\OpticalSystemOffsetAdjustment

Check the latest text.log files as shown in the Figure 23.

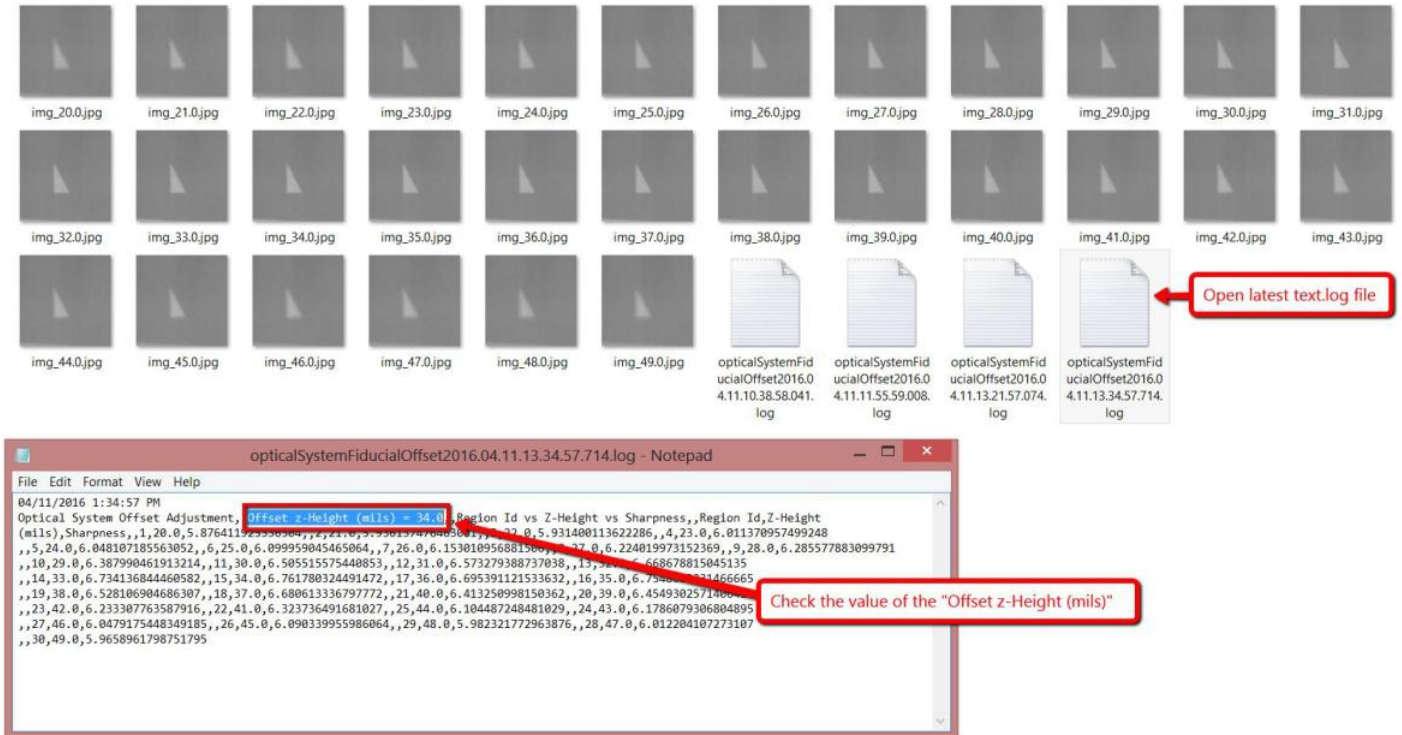


Figure 23 Z-Offset

Check the value of the Offset z-Height (mils) whether it is within the range listed in Table 4

System Type	Low Limit	High Limit
Standard	-95 mils	- 75 mils

Table 4 Z-Offset Limit

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21. If the Optical System Offset Adjustment failed as shown in Figure 24, go to the following folder
Directory: *C:\Program Files\AXI System\v810\5.0\log\calib\OpticalSystemOffsetAdjustment* to
detect the failure by refer to Figure 25 & 26.



Figure 24 Optical System Offset Adjustment

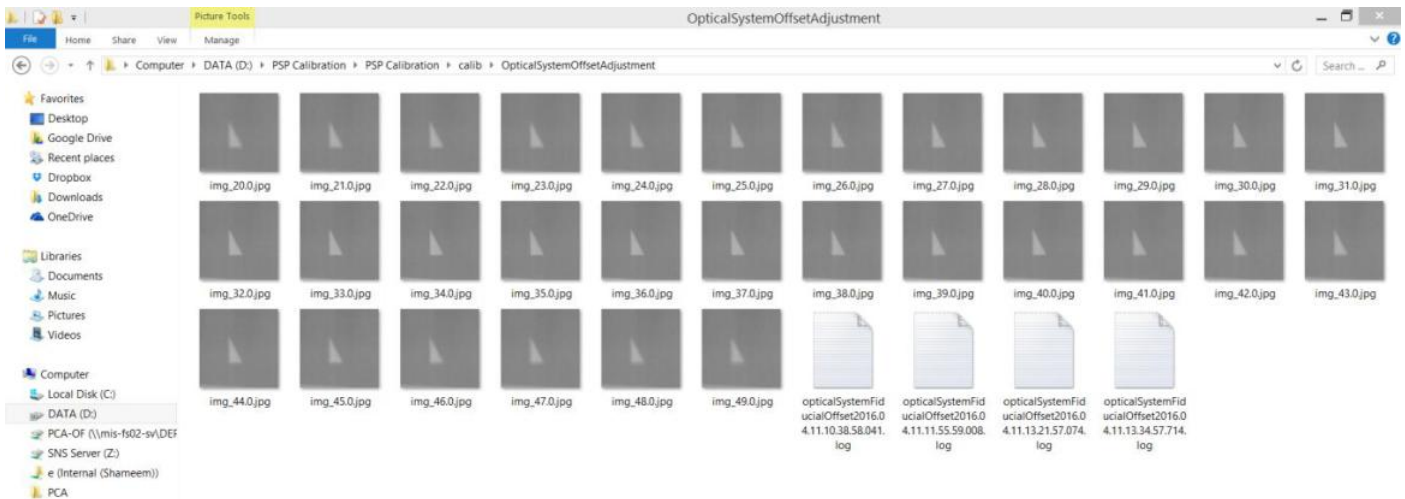


Figure 25 Good Images

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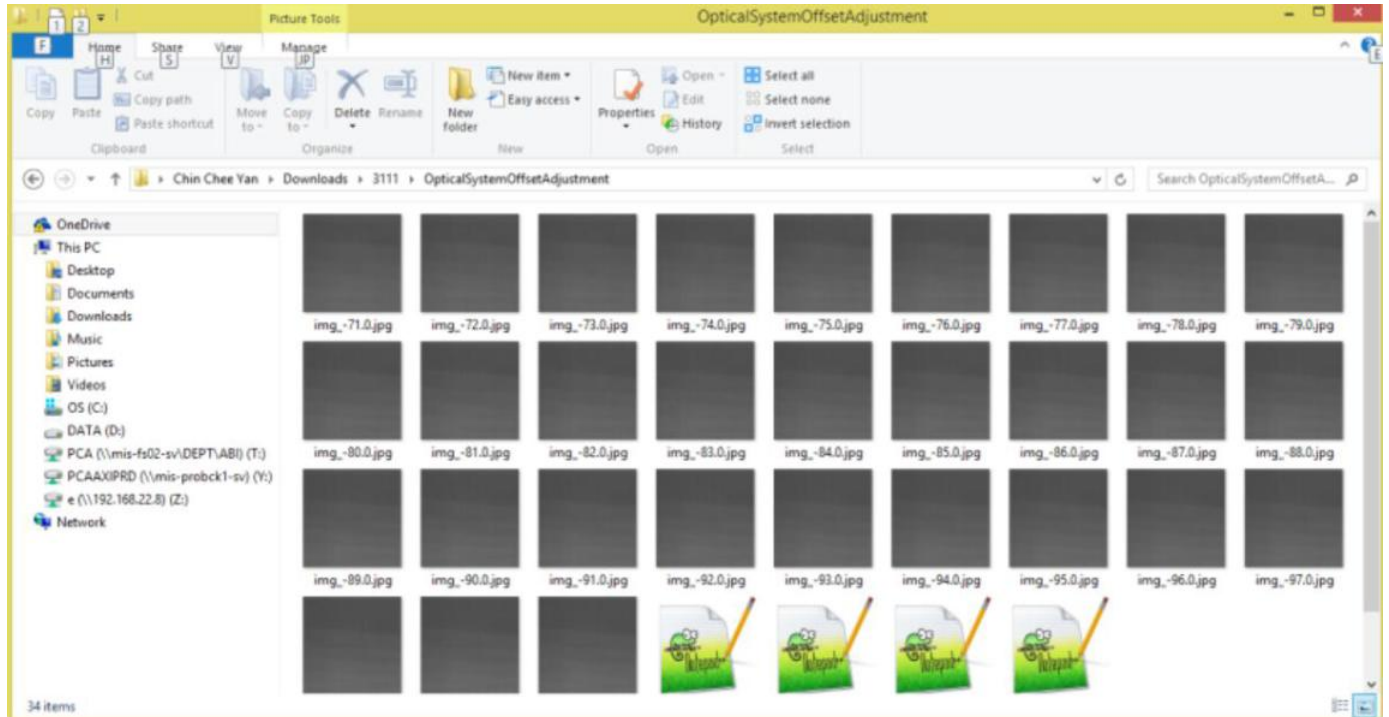


Figure 26 Bad Images

If the images generated are as in Figure 26, check the following items as below.

- i. Require to check the existence of Fiducial Coupon which installed at bottom part of the On rail jig.



Figure 27 Fiducial Coupon

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ii. Visual check existence of On rail jig at the adjustable rail as shown in the Figure 28.

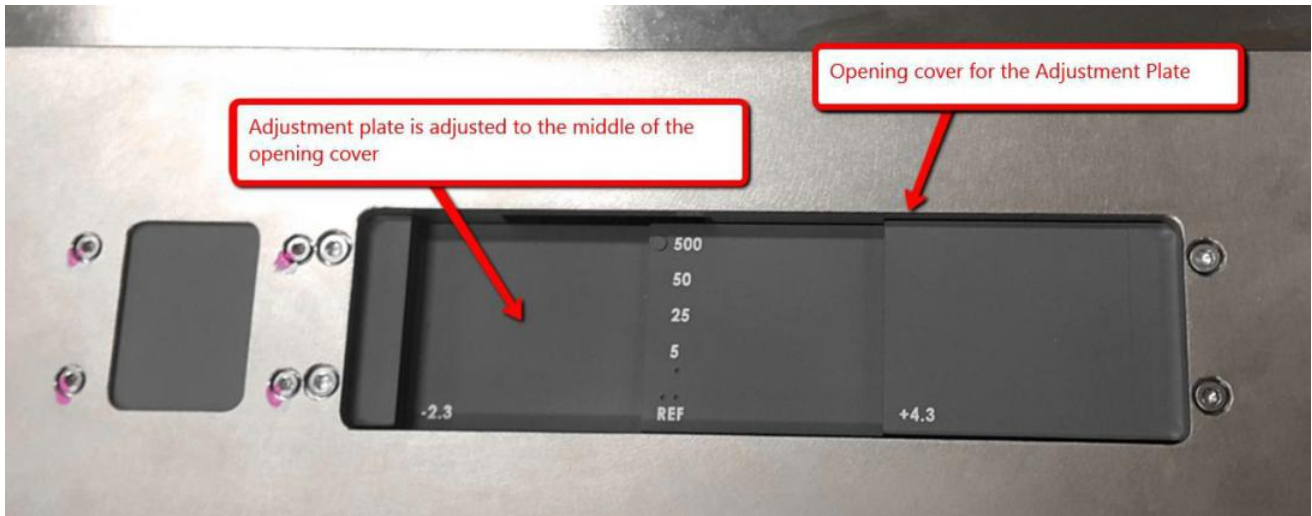


Figure 28 Adjustment Plate adjustment

22. At **Confirmation, Diagnostics and Adjustment (CD&A)**, under the **Manual Optical Confirmation**, right click **Front Optical Measurement Repeatability Confirmation** and select **Run test without dependent tests** as shown in the Figure 29.

Remarks: The first run will mainly for 500 micron test.

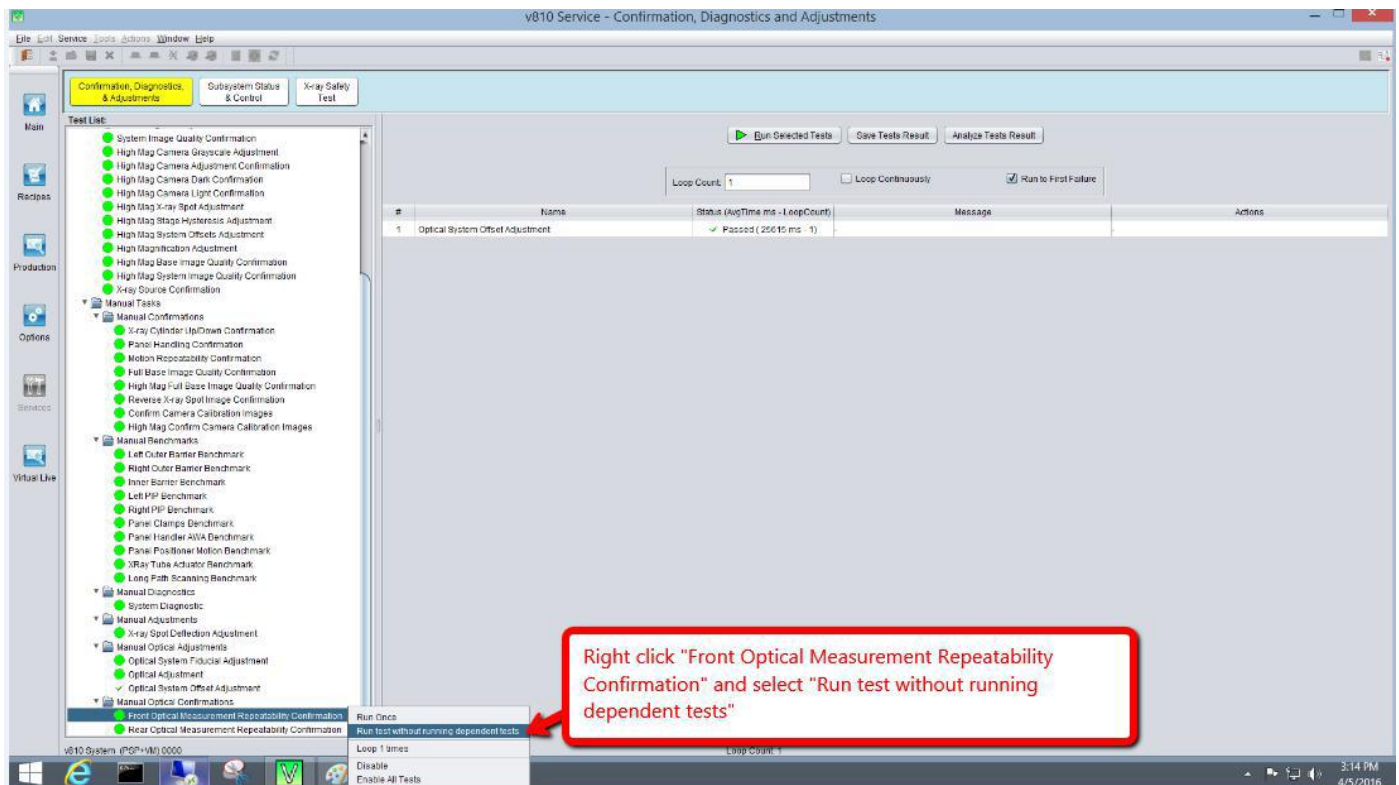


Figure 29 Front Optical Measurement Repeatability Confirmation

23. Select the points as indicated in Figure 30. Total of 6 points require to be select as shown in the Figure 30.

Remarks: Click to choose the point.

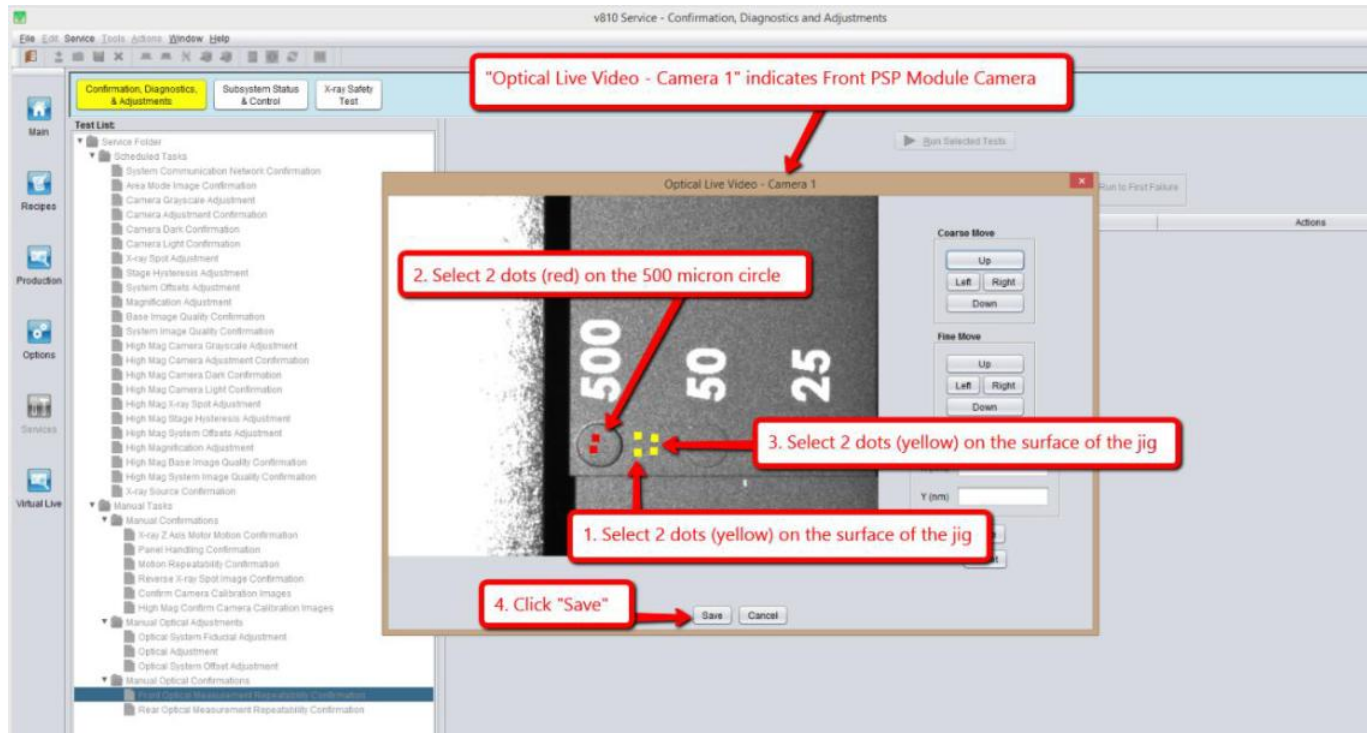


Figure 30 Adjustment Points

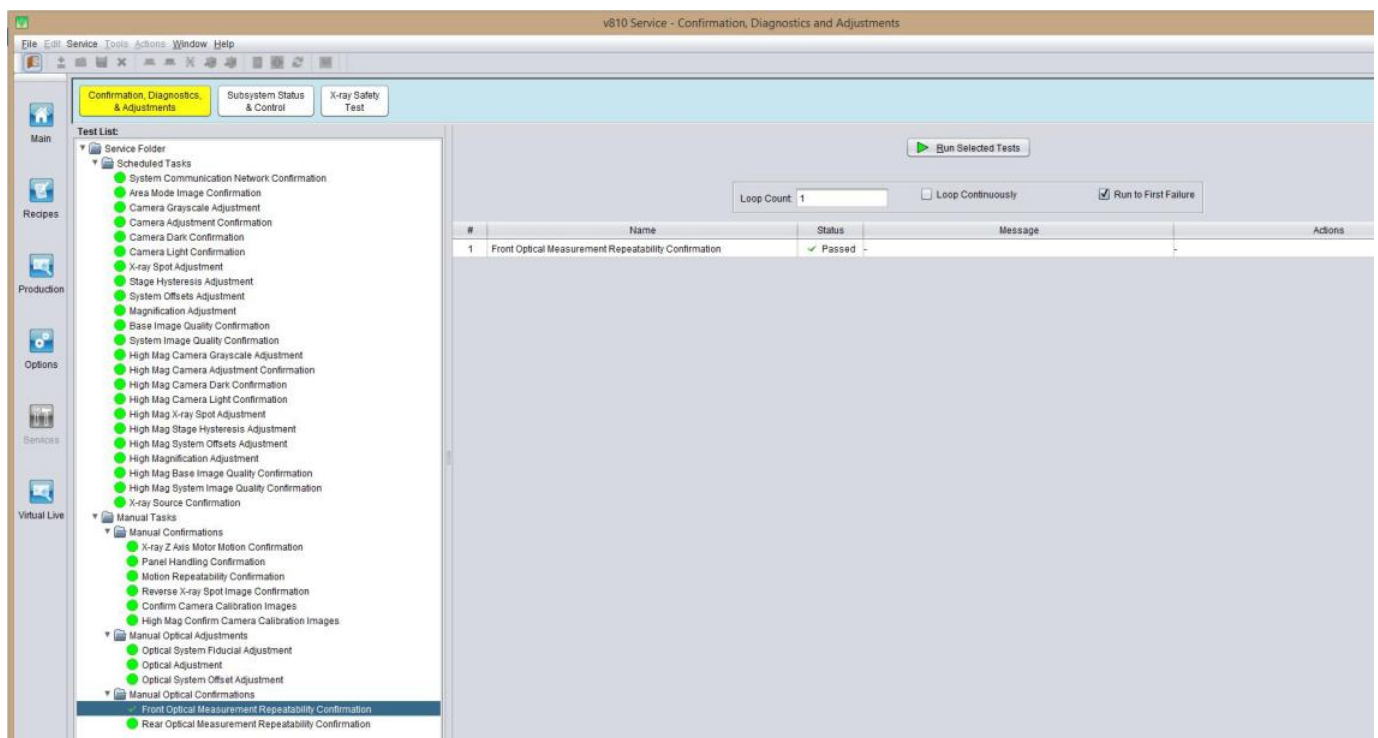


Figure 31 Front Optical Measurement Repeatability Confirmation Passed

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25. At **Confirmation, Diagnostics and Adjustment (CD&A)**, under the **Manual Optical Confirmation**, right click **Front Optical Measurement Repeatability Confirmation** and select **Run test without dependent tests** as shown in the Figure 33.

Remarks: The 2nd run will mainly for 50 micron test.

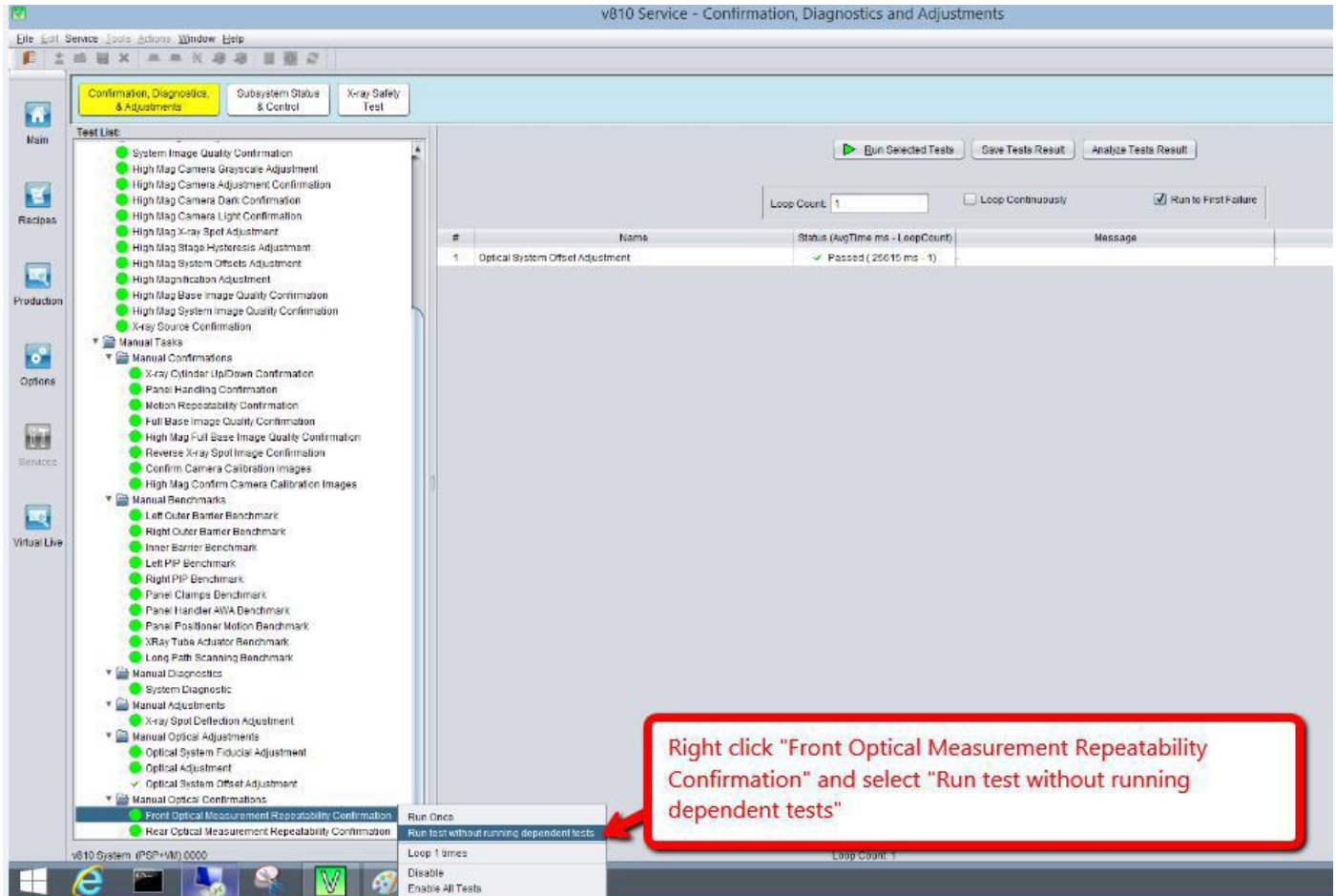


Figure 33 Front Optical Measurement Repeatability Confirmation

26. Select particular points as indicated in Figure 34. Total of 6 points require to be select as shown in the Figure 34.

Remarks: Click to choose the point.

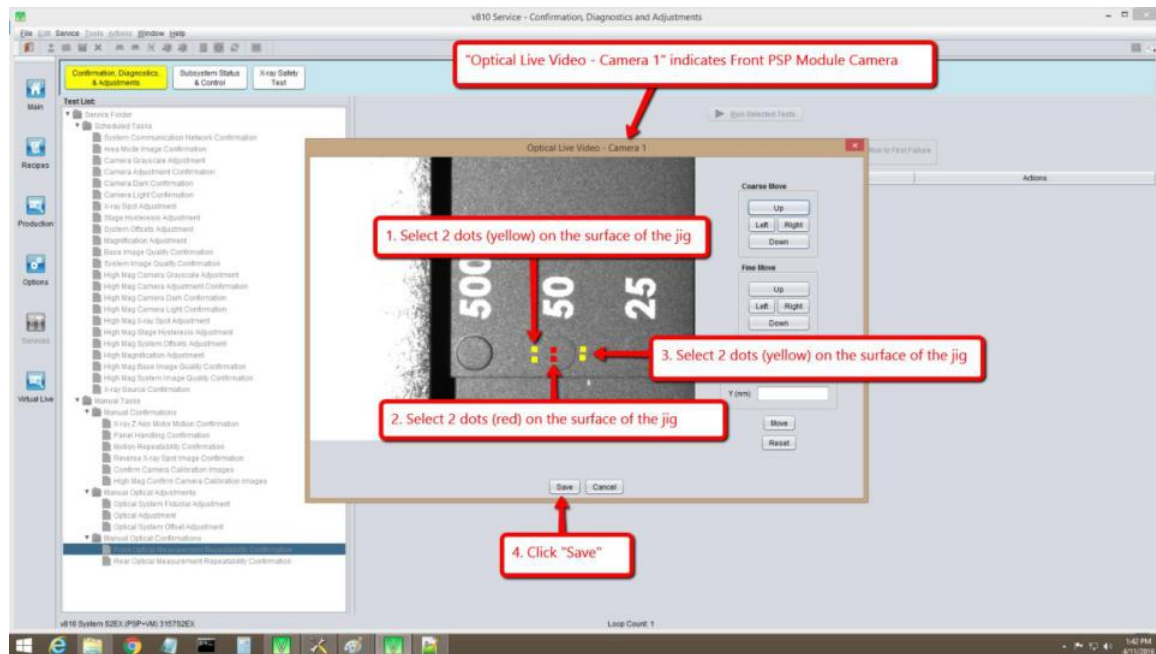


Figure 34 Adjustment Points

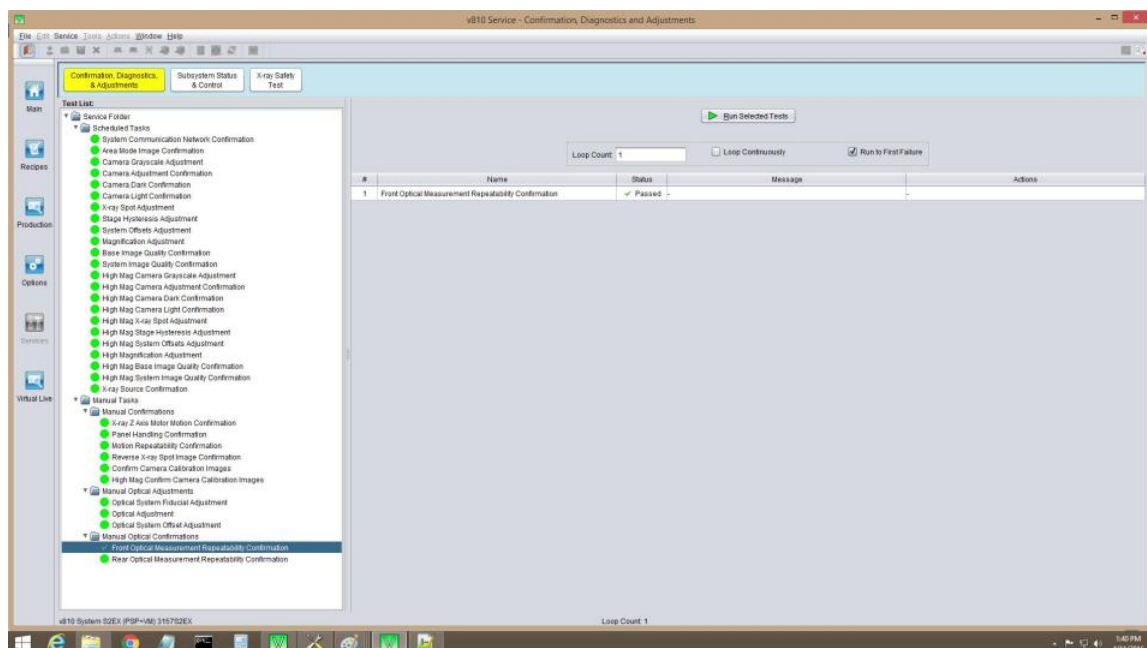


Figure 35 Front Optical Measurement Repeatability Confirmation Passed

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27. Go to the following directory.

Directory: C:\Program Files\AXI

System\v810\5.0\log\confirm\FrontOpticalMeasurementRepeatability

Open the latest text.log file as shown in the Figure 36 and check the value of the “Mean Z-Height (nm)” and “StdDev Z-Height (nm)” whether those values are within the range listed in Table 6.

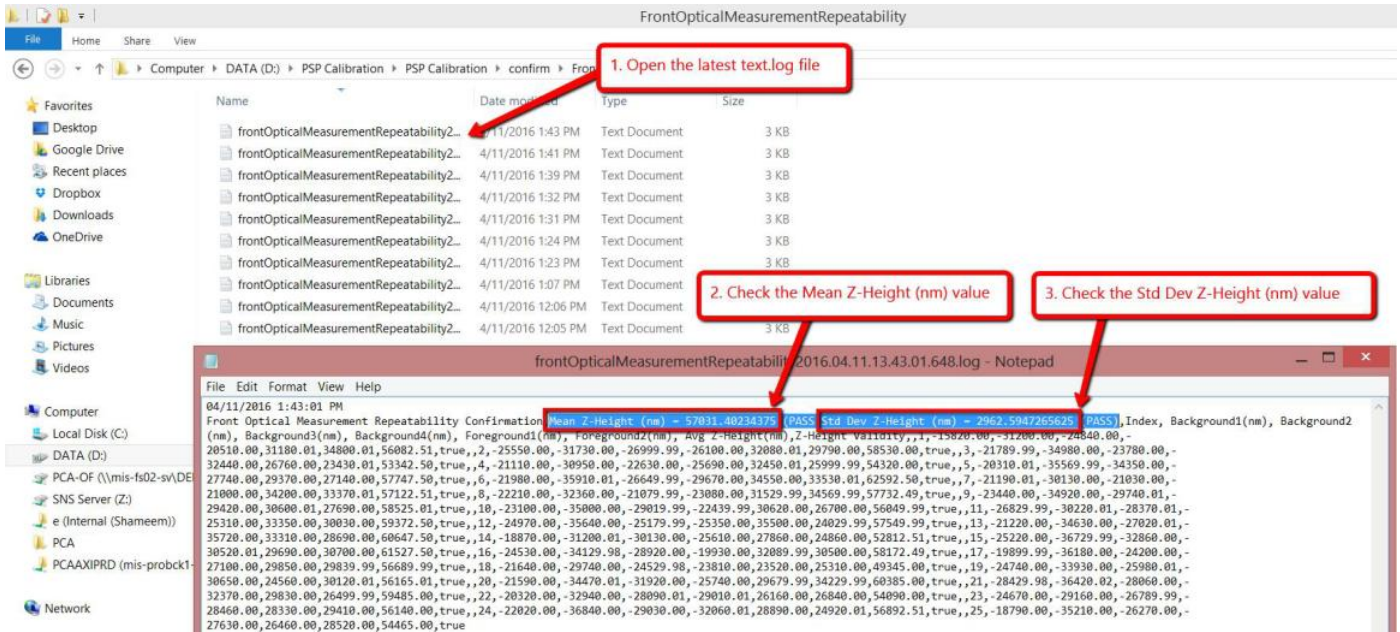


Figure 36 Repeatability test

Log files	Low Limit	High Limit
Mean Z-Height (nm)	20000.0	80000.0
StdDev Z-Height (nm)	0.0	5000.0

Table 6 50 Micron repeatability test

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28. At **Confirmation, Diagnostics and Adjustment (CD&A)**, under the **Manual Optical Confirmation**, right click **Rear Optical Measurement Repeatability Confirmation** and select **Run test without dependent tests** as shown in the Figure 37.

Remarks: The first run will mainly for 500 micron test.

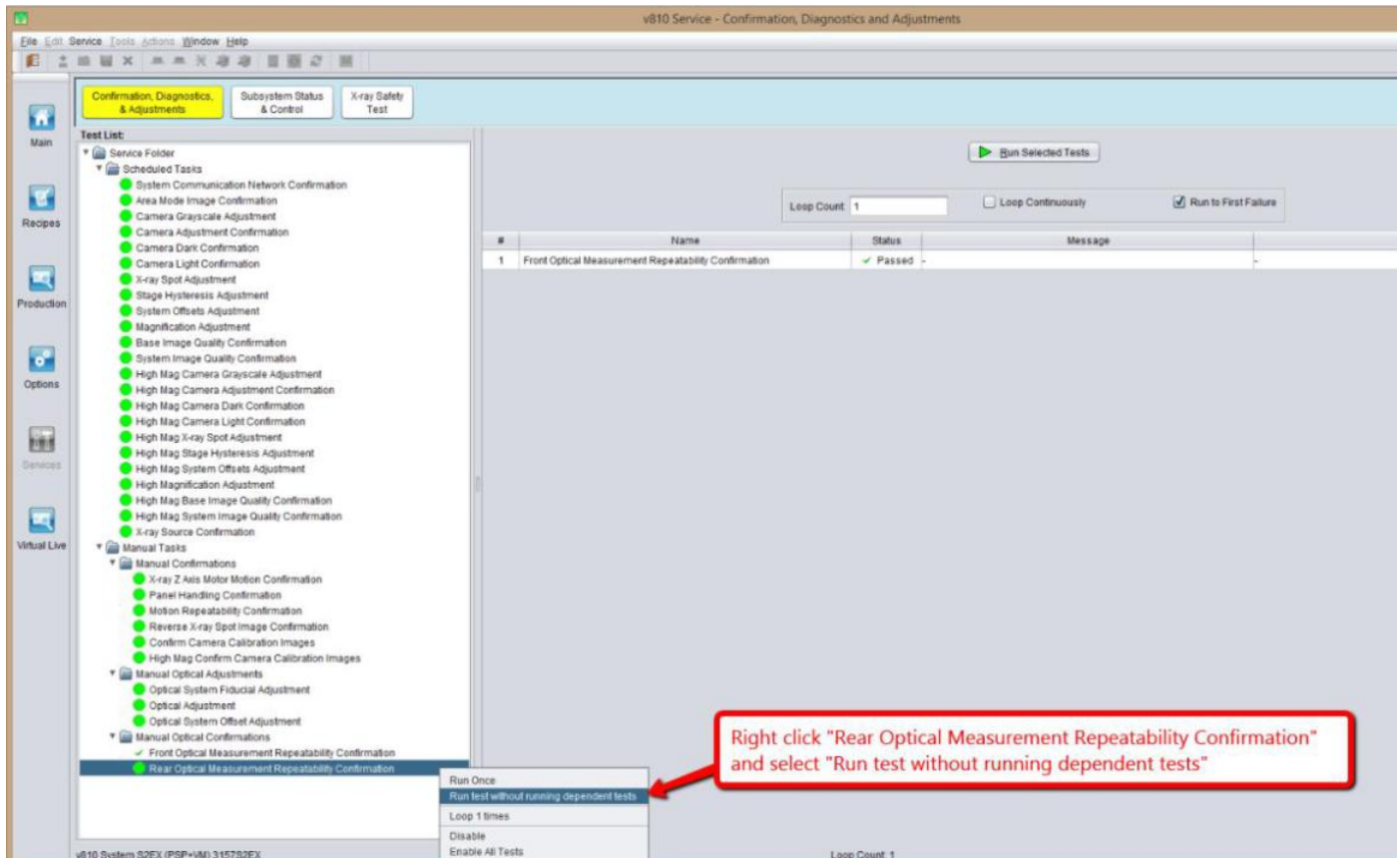


Figure 37 Rear Optical Measurement Repeatability Confirmation

29. Select the points as indicated in Figure 38. Total of 6 points require to be select as shown in the Figure 38.

Remarks: Click to choose the point.

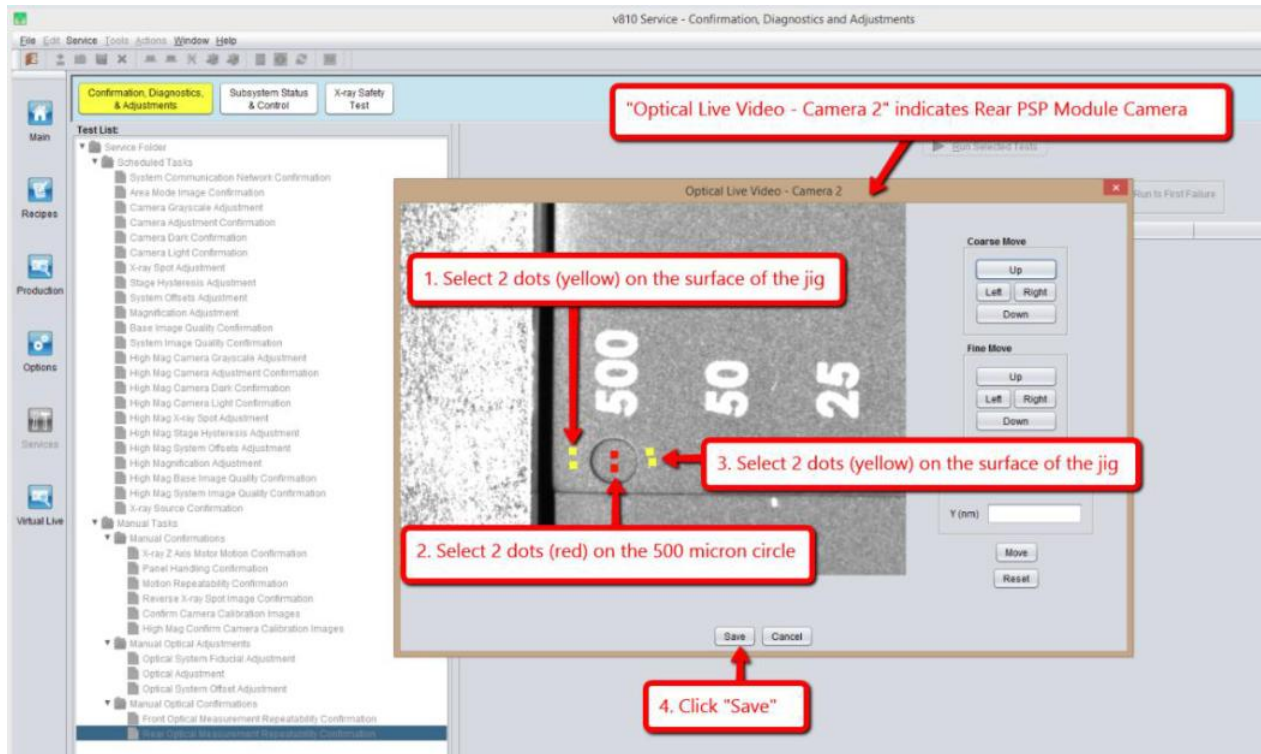


Figure 38 Adjustment Points

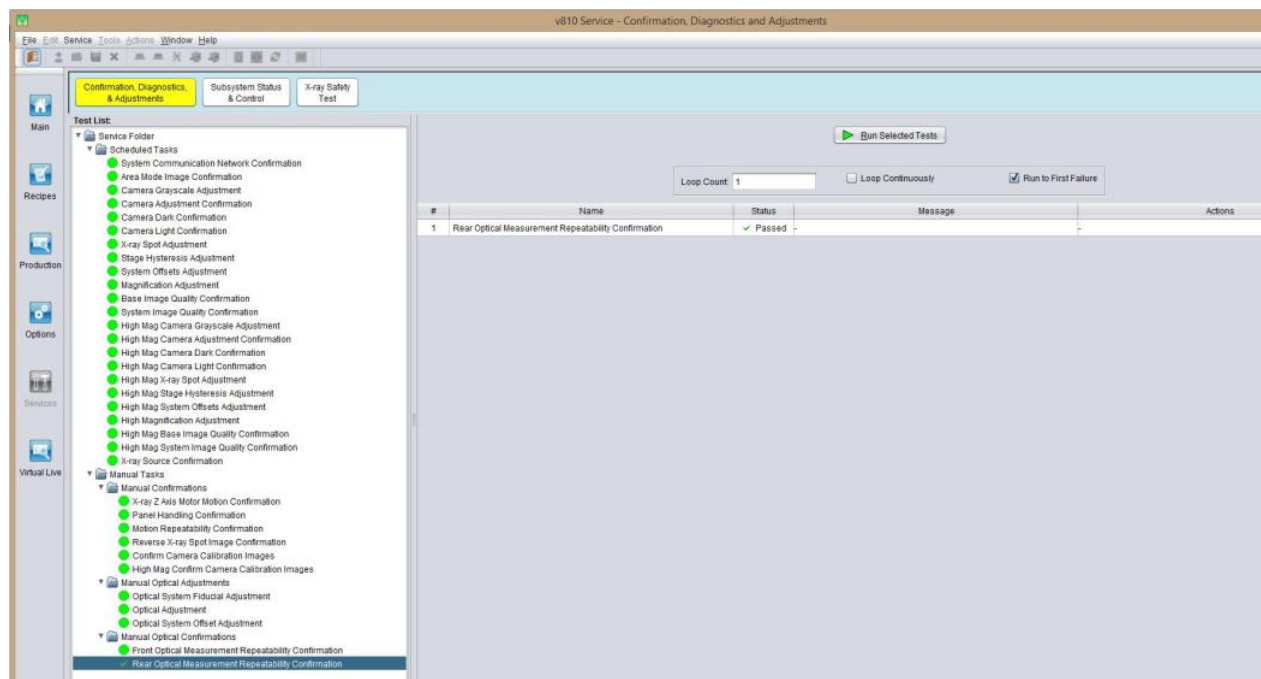


Figure 39 Rear Optical Measurement Repeatability Confirmation Passed

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30. Go to the following directory.

Directory: C:\Program Files\AXI

System\v810\5.0\log\confirm\RearOpticalMeasurementRepeatability

Open the latest text.log file as shown in the Figure 40 and check the value of the “Mean Z-Height (nm)” and “StdDev Z-Height (nm)” whether those values are within the range listed in Table 7.

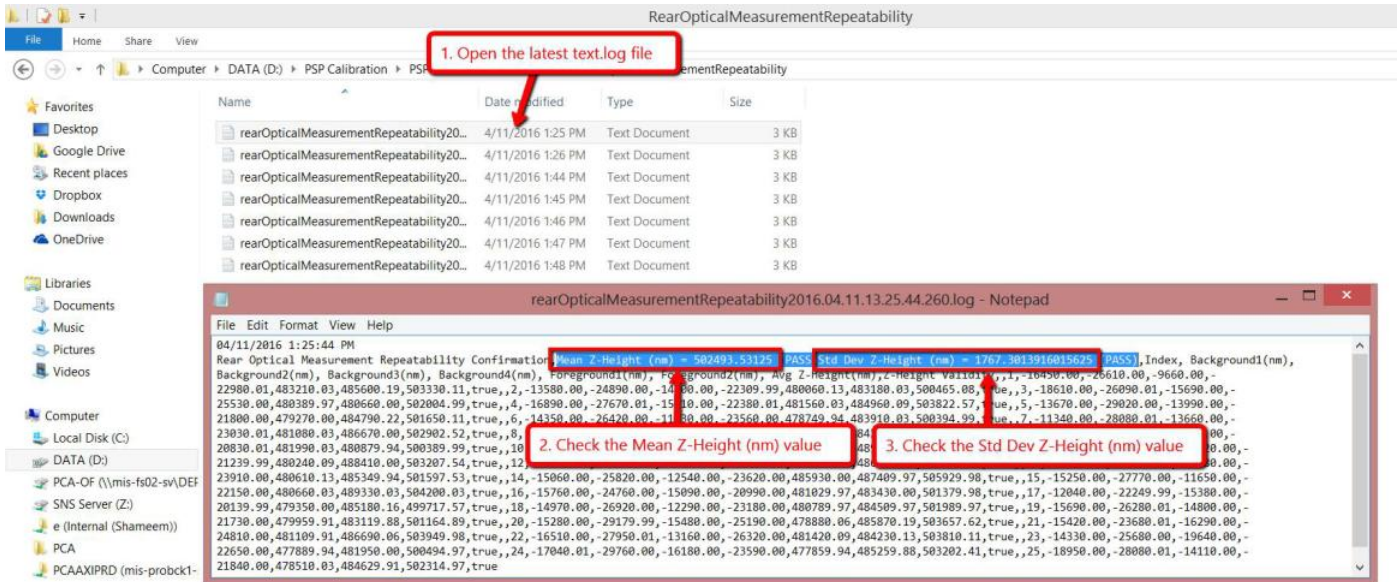


Figure 40 Repeatability test

Log files	Low Limit	High Limit
Mean Z-Height (nm)	400000.0	600000.0
StdDev Z-Height (nm)	0.0	5000.0

Table 7 500 Micron repeatability test

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31. At **Confirmation, Diagnostics and Adjustment (CD&A)**, under the **Manual Optical Confirmation**, right click **Front Optical Measurement Repeatability Confirmation** and select **Run test without dependent tests** as shown in the Figure 41.

Remarks: The second run will mainly for 50 micron test.

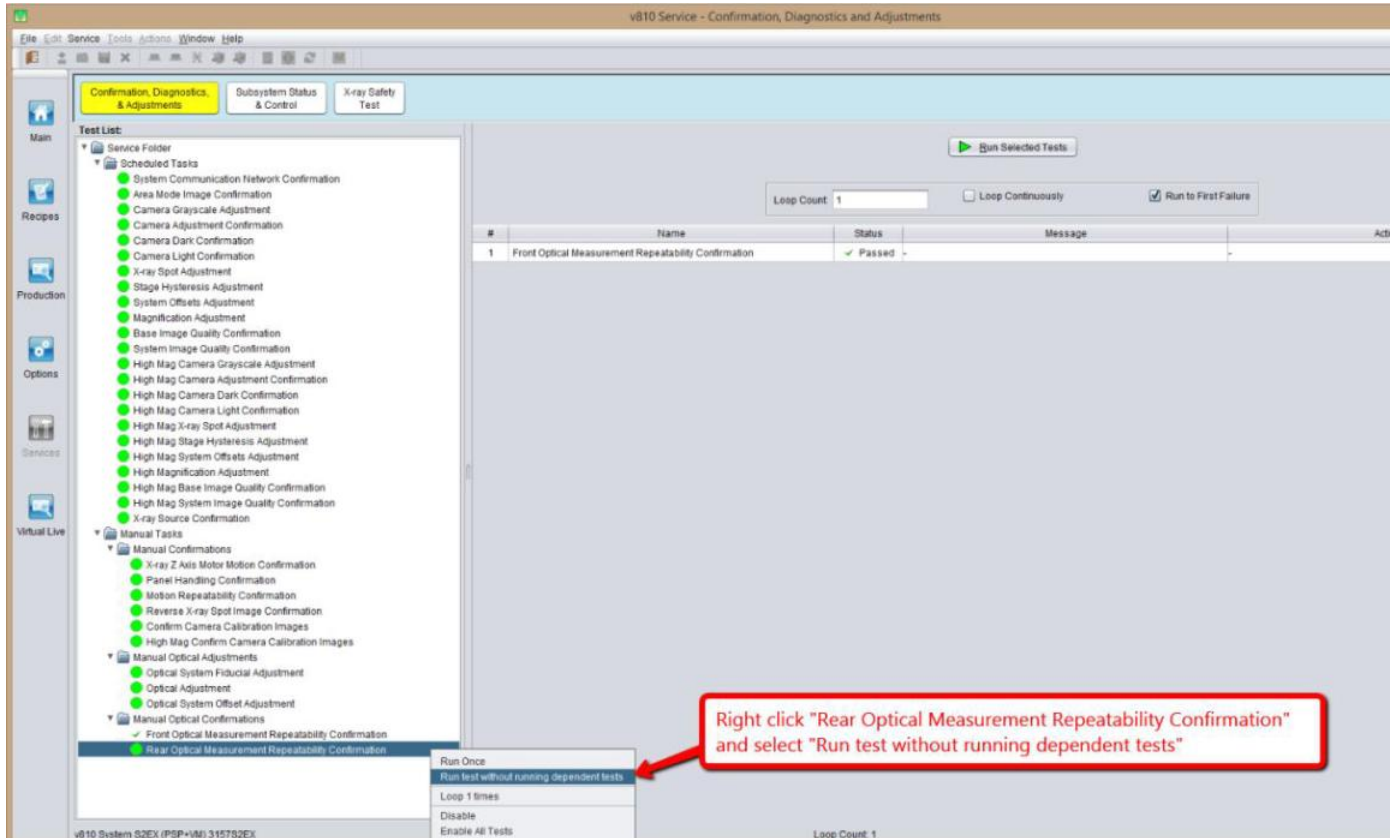


Figure 41 Front Optical Measurement Repeatability Confirmation

32. Select particular points as indicated in Figure 42. Total of 6 points require to be select as shown in the Figure 42.

Remarks: Click to choose the point.

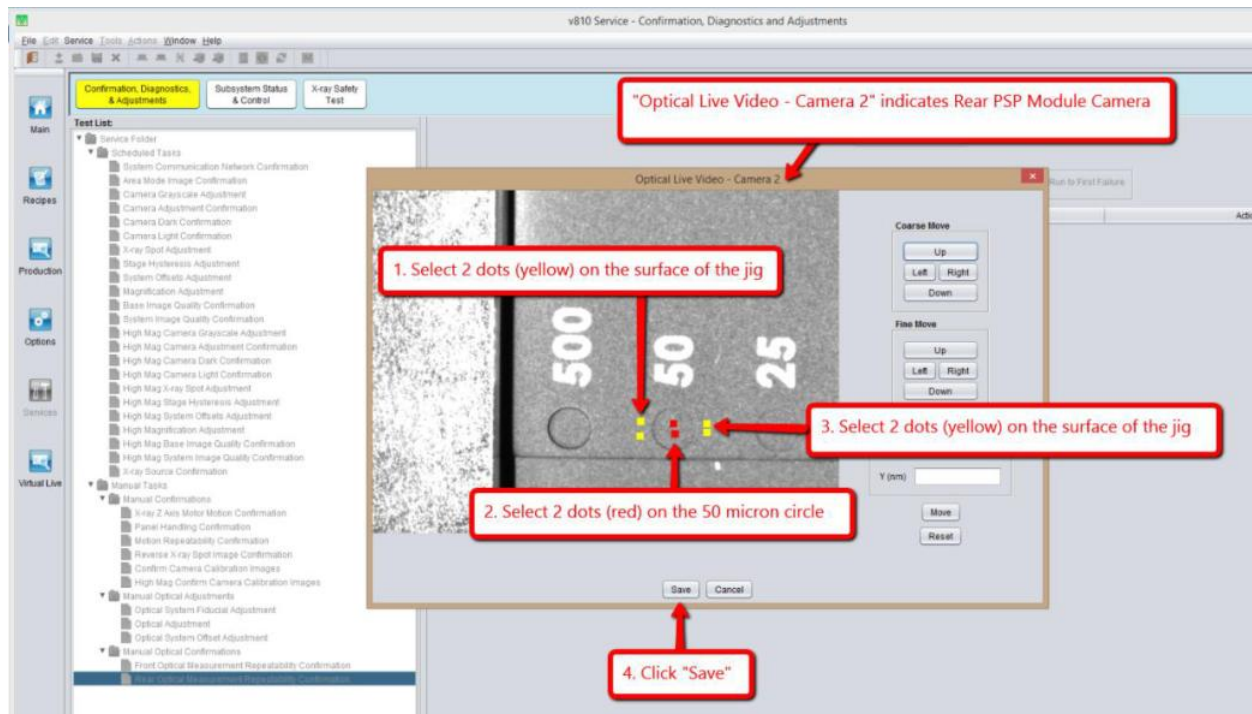


Figure 42 Adjustment Points

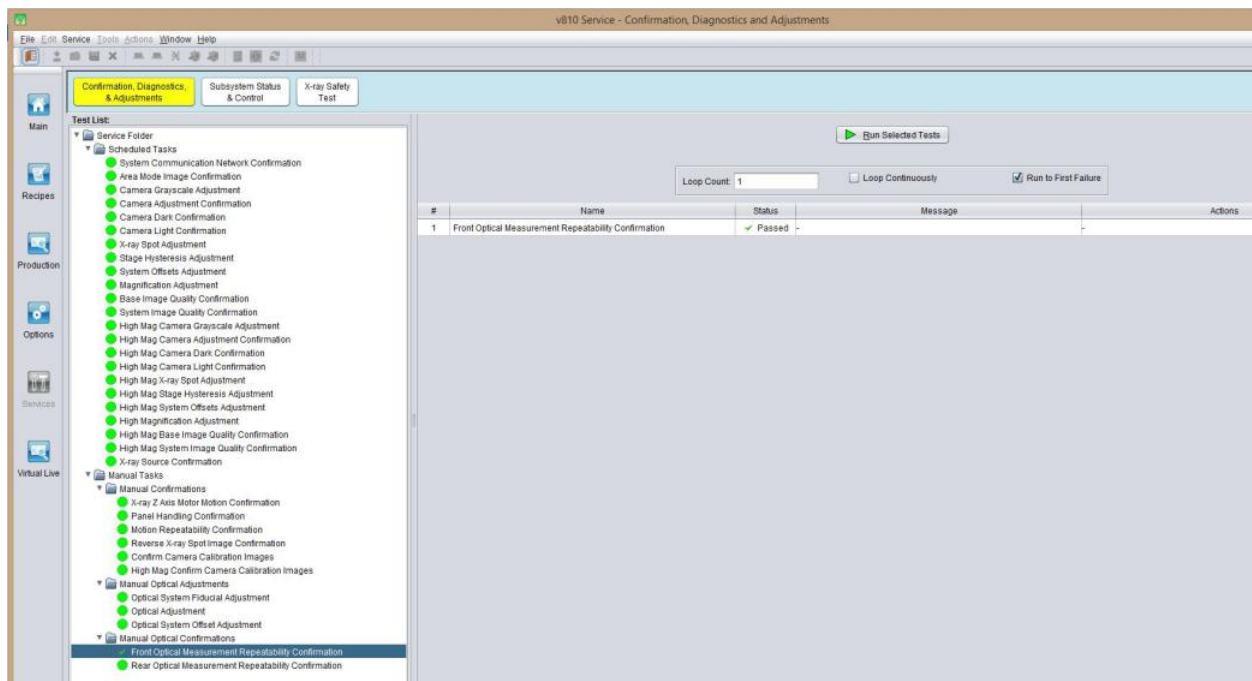


Figure 43 Front Optical Measurement Repeatability Confirmation Passed

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33. Go to the following directory.

Directory: C:\Program Files\AXI

System\v810\5.0\log\confirm\RearOpticalMeasurementRepeatability

Open the latest text.log file as shown in the Figure 44 and check the value of the “Mean Z-Height (nm)” and “StdDev Z-Height (nm)” whether those values are within the range listed in Table 8.

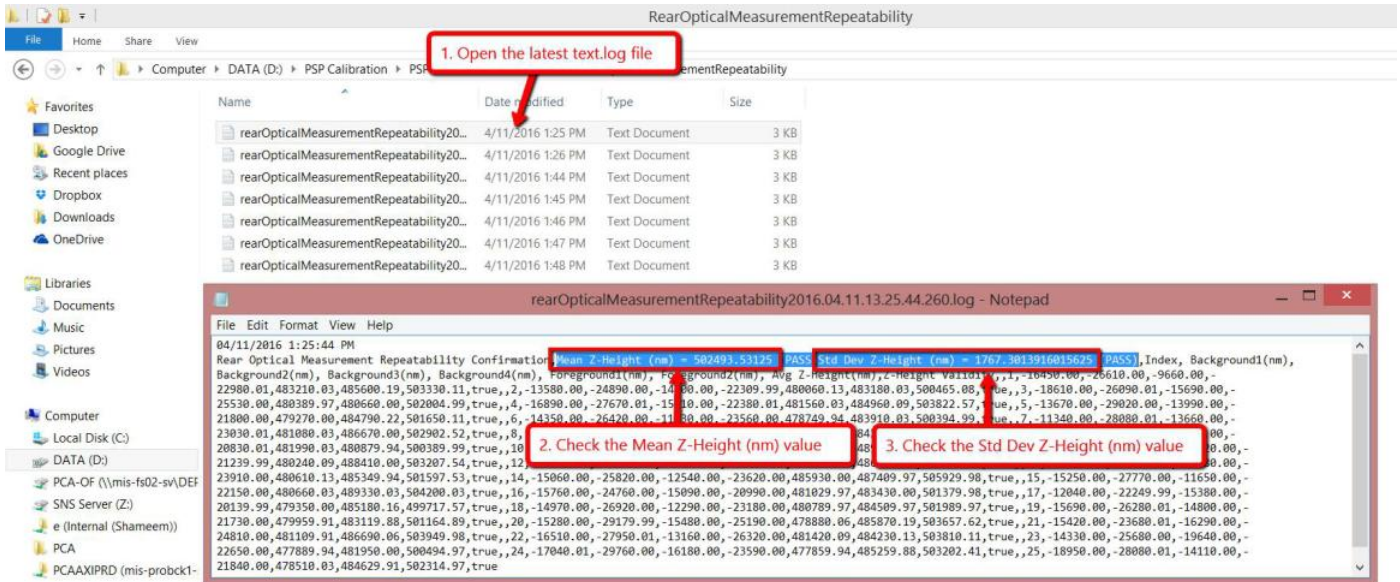


Figure 44 Repeatability test

Log files	Low Limit	High Limit
Mean Z-Height (nm)	20000.0	80000.0
StdDev Z-Height (nm)	0.0	5000.0

Table 8 50 Micron repeatability test

34. Send below folder to Vitrox AXI SNS for further verification

i. OpticalCalibration (whole folder)

Directory: C:\Program Files\AXI System\v810\5.0\log\calib

ii. OpticalSystemOffsetAdjustment (whole folder)

Directory: C:\Program Files\AXI System\v810\5.0\log\calib

iii. FrontOpticalMeasurementRepeatability (whole folder)

Directory: C:\Program Files\AXI System\v810\5.0\log\confirm

iv. RearOpticalMeasurementRepeatability (whole folder)

Directory: C:\Program Files\AXI System\v810\5.0\log\confirm